Sensing and the Imaging Process (II)

Introduction to Computational and Biological Vision

CS 202-1-5261

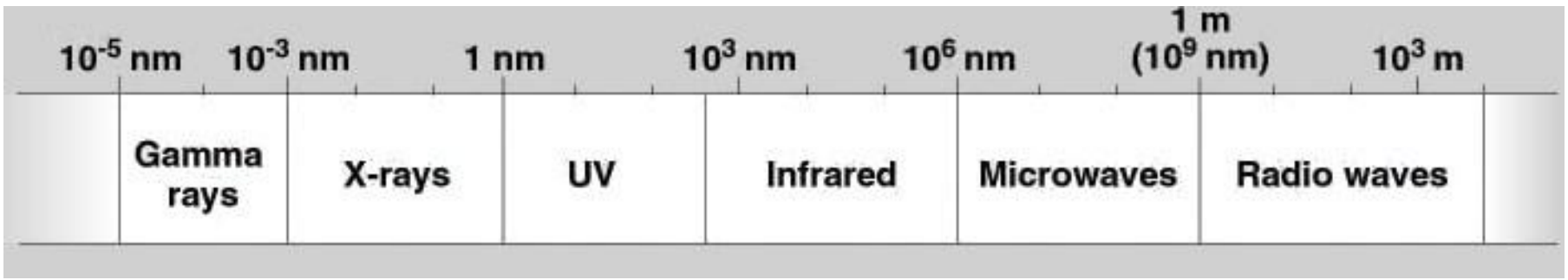
Computer Science Department, BGU

Ohad Ben-Shahar

What is light

The electromagnetic spectrum

Shorter Wavelength → **Longer Wavelength**



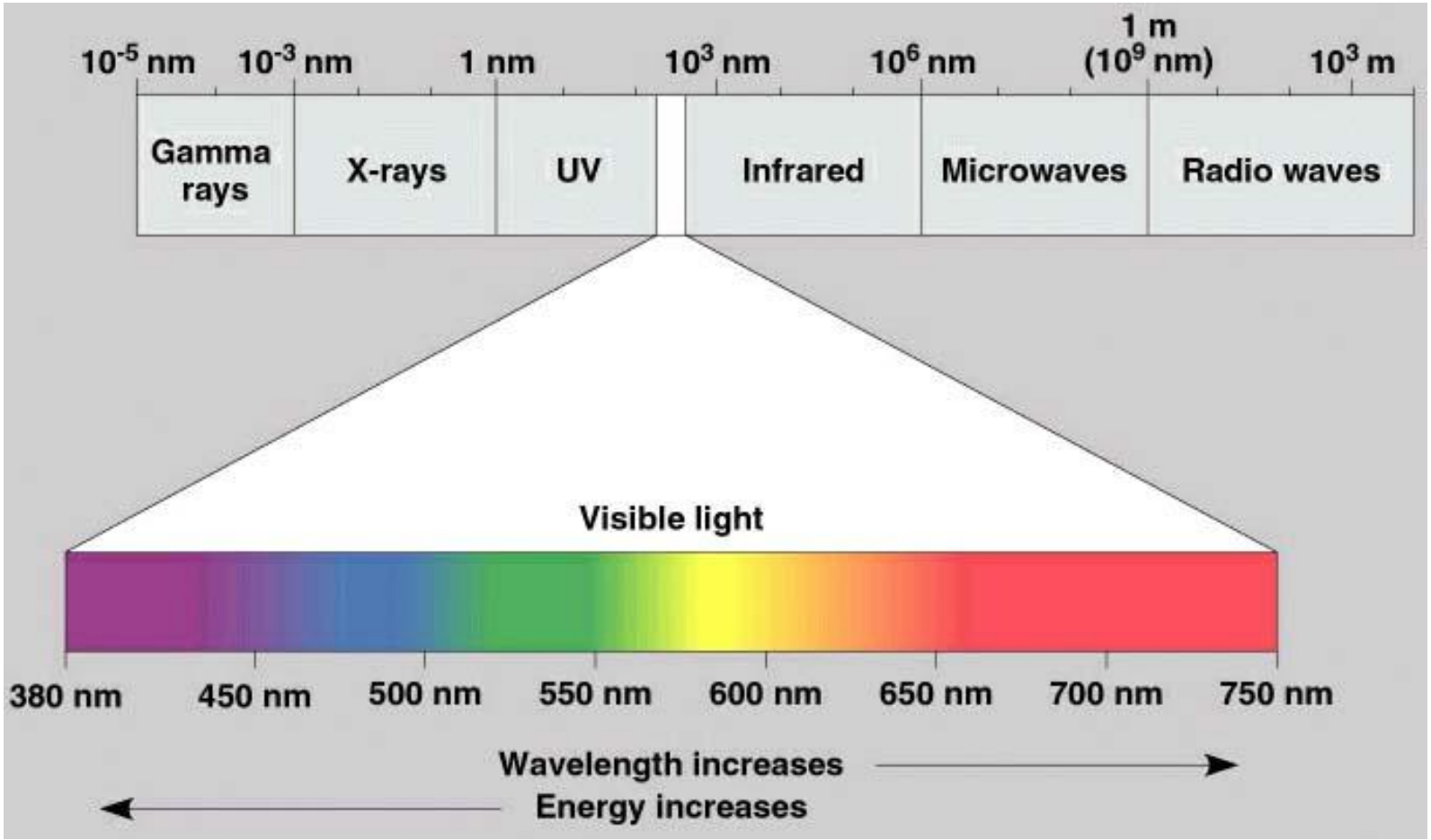
← **Higher frequency**
Higher energy

Smaller frequency
Smaller energy

$$f = \frac{c}{\lambda} \qquad c = 2.998 \cdot 10^{8} m/s$$

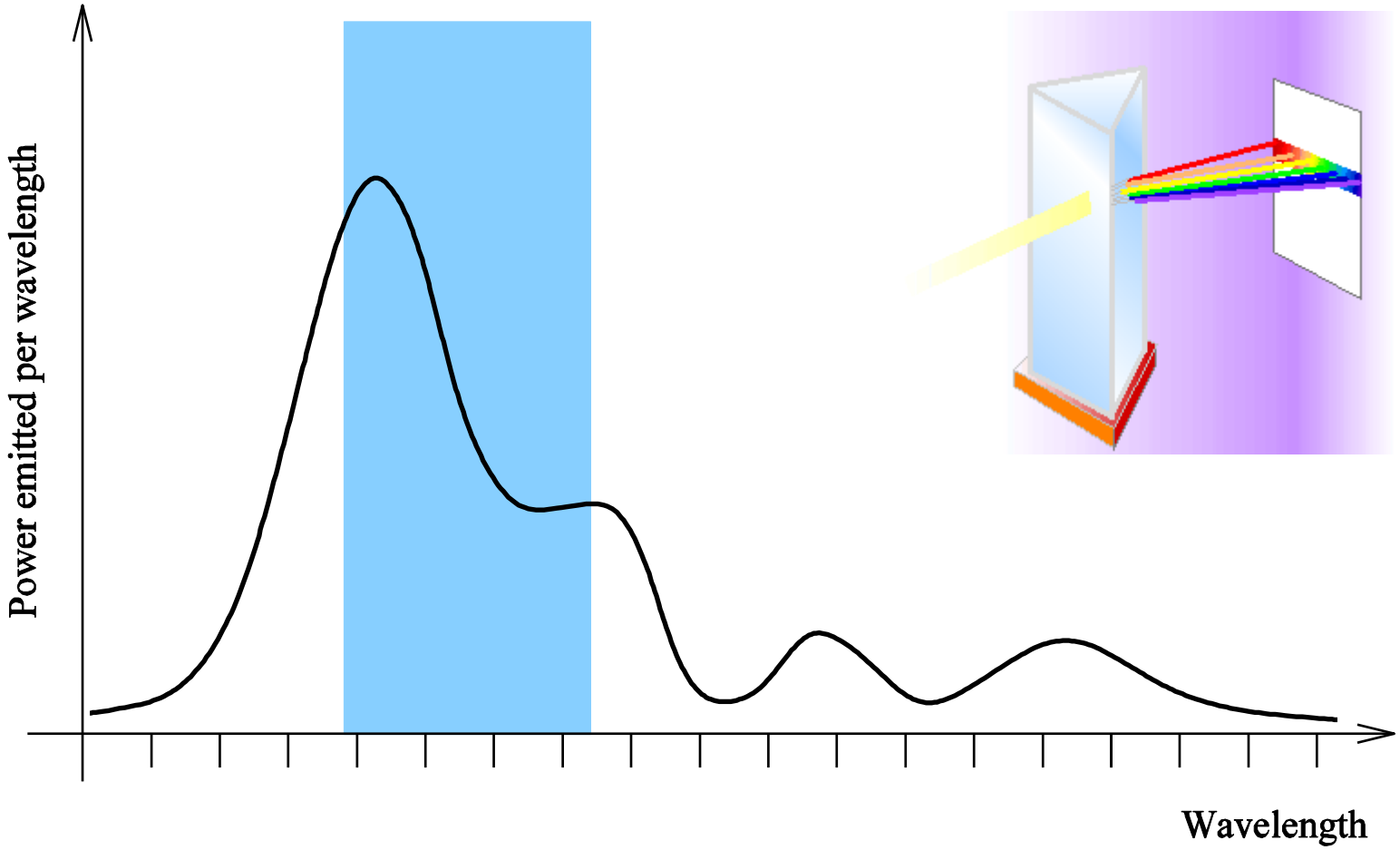
$$E = h \cdot f \qquad h = 6.623 \cdot 10^{34} Js$$

What is light



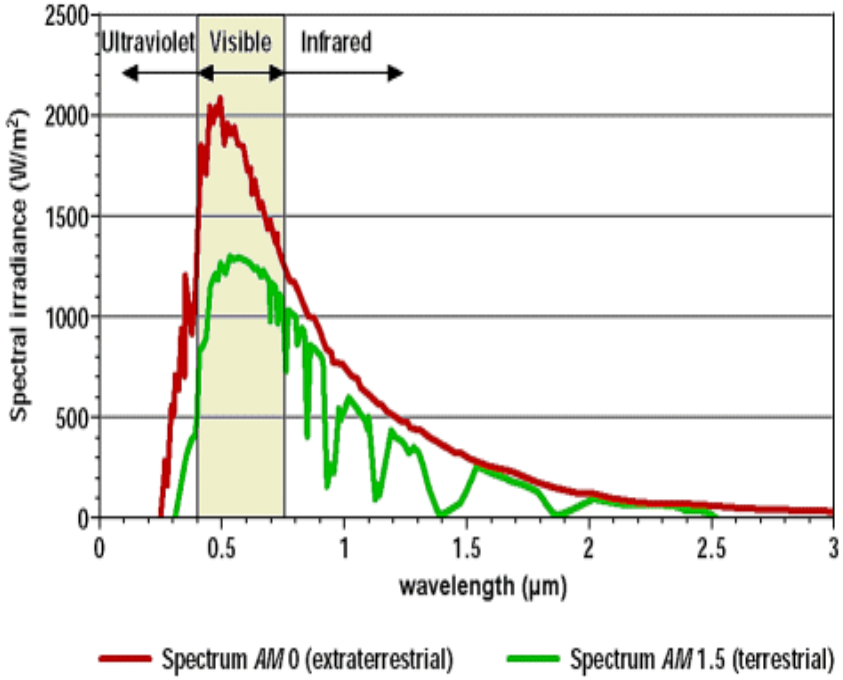
What is light

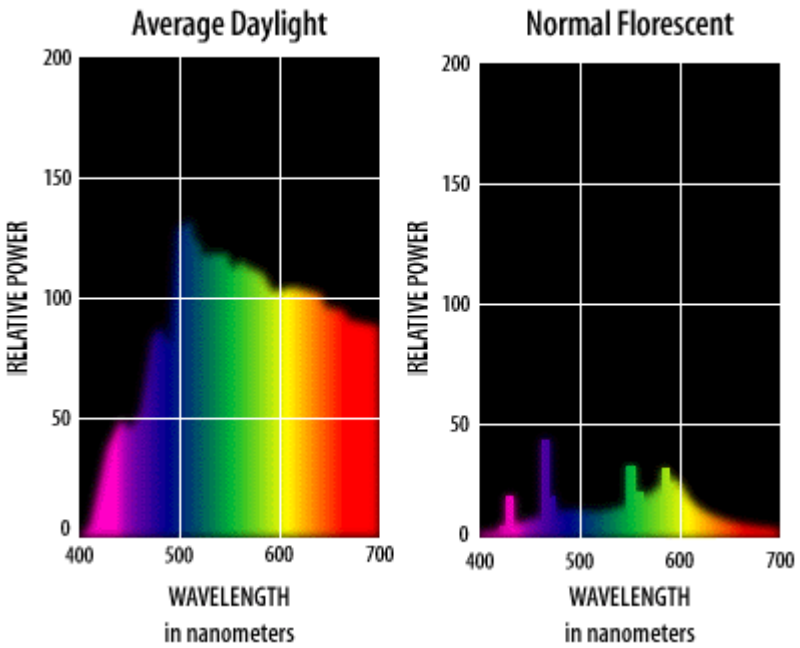
Power spectrum (spectral power)



What is light

A light source can be described by its spectral power





Should we expect biological light-capturing devices to evolve?

"The sun …offered the chance of remote guidance technology. It pummeled every square millimeter of Earth's surface with a fusillade of photos: tiny particle allows, criss-cr ook escaped aight lines and so ers and reflecte e always been so ty for remote s necessary only to s from which they came. Would the opportunity be taken up?

Three billion years later you know the answer, for you can see these words."

"בגלל שפוטונים נעים במסלולים ישרים ובמהירות עצומה, בגלל שהם נבלעים בחומרים מסויימים יותר מאשר באחרים ומוחזרים מחומרים מסויימים יותר מאשר מאחרים, ובגלל שהם כה רבים ואינם מתכלים, הפוטונים סיפקו [לאבולוציה] הזדמנות לטכנולוגית חישה מרחוק בעלת עוצמה ודיוק כבירים."

Richard Dawkins, *The Forty-Fold Path to Enlightenment*, 1996

What is an eye

"Eyes"

Camera eyes

Formation of Spatial Vision

how much light is coming from different directions,

or… an image is formed

Pigment pits

(Ocelli, Stemmata, Ommatidia)

What is an eye

~540 MYA

(during Cambrian explosion)

>3 BYA

(pigment evolution)

"Eyes"

Camera eyes

Formation of Spatial Vision

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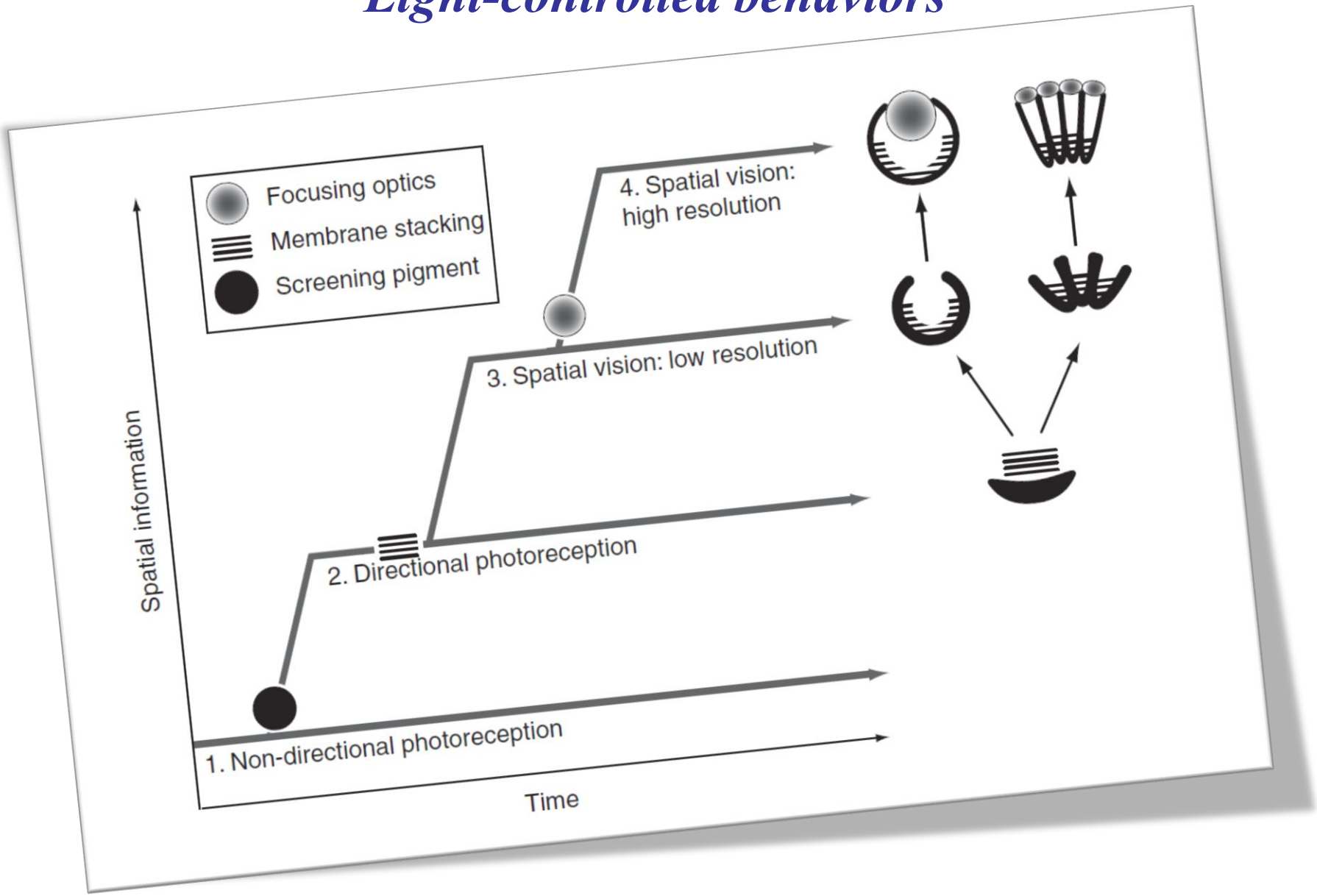
Pigment pits

(Ocelli, Stemmata, Ommatidia)

Light-controlled behaviors



Light-controlled behaviors

1. **Behaviors controlled by non-directional monitoring of ambient light**
   - circadian rhythms
   - light-avoidance responses for protection against harmful levels of short wavelength light
   - Shadow responses to avoid predation
2. **Behaviors based on rough (say 2) directional light sensitivity**
   - phototaxis
   - control of body posture (optical balance)
   - alarm responses for approaching predators
3. **Visual tasks based on low spatial resolution**
   - object avoidance responses (anti-collision)
   - habitat selection
   - orientation to coarse landmarks or major celestial objects
4. **Visual tasks based on high spatial resolution**
   - detection and pursuit of prey
   - predator detection and evasion
   - mate detection and evaluation
   - visual communication
   - Recognition of individuals

What is an eye

"Eyes"

Camera eyes

Formation of Spatial Vision

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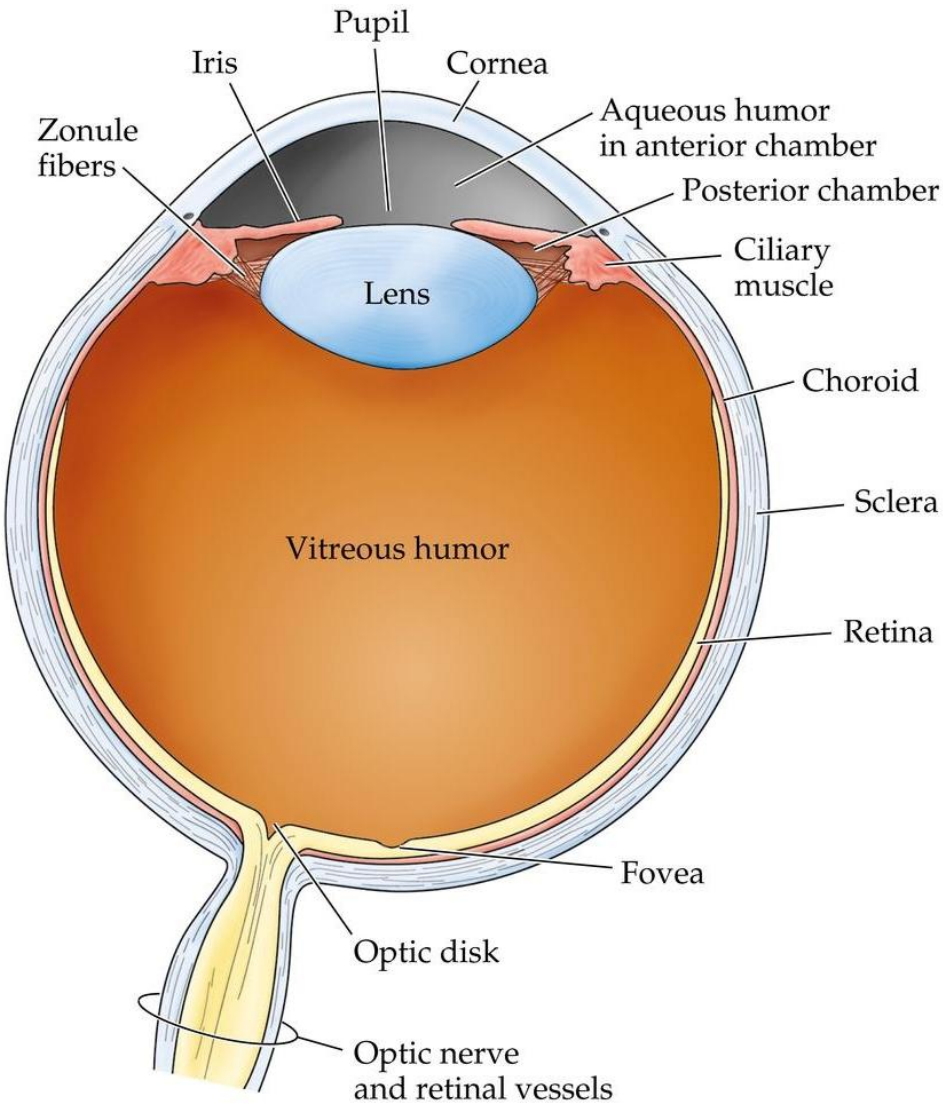
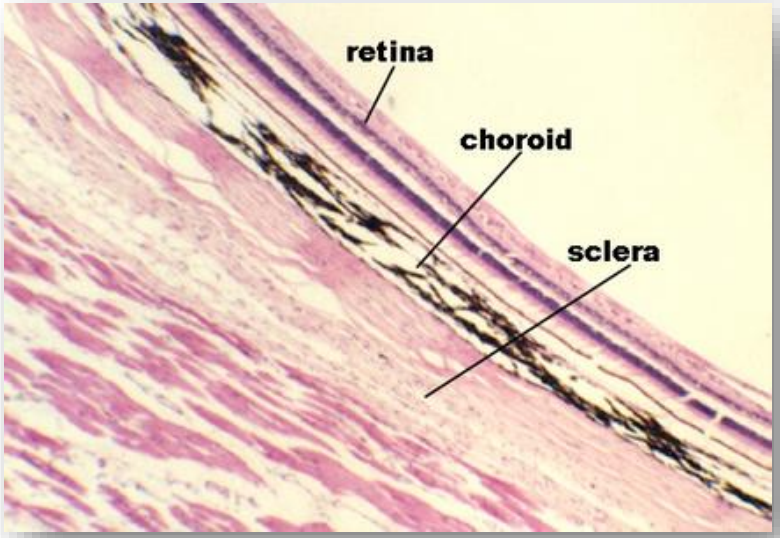
Pigment pits

(Ocelli, Stemmata, Ommatidia)

The human eye

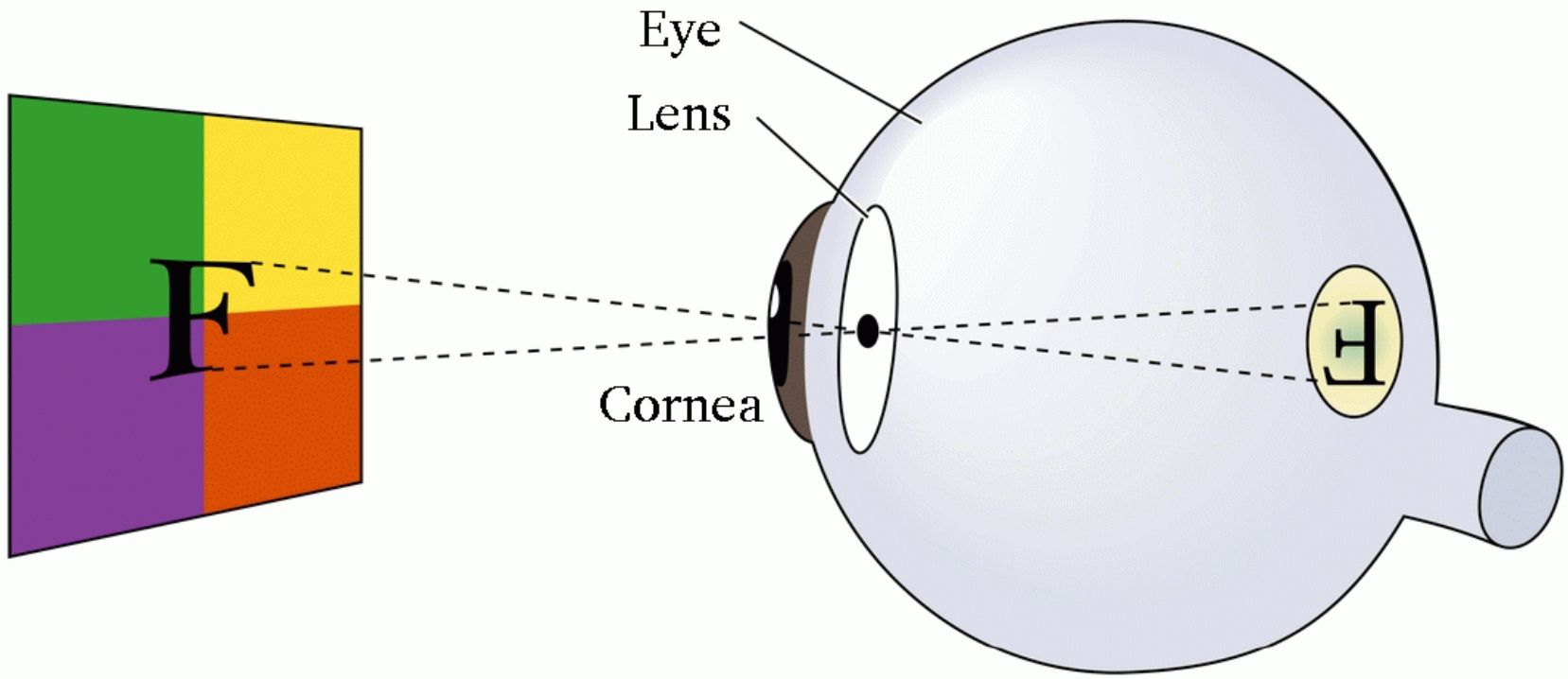
The human eye

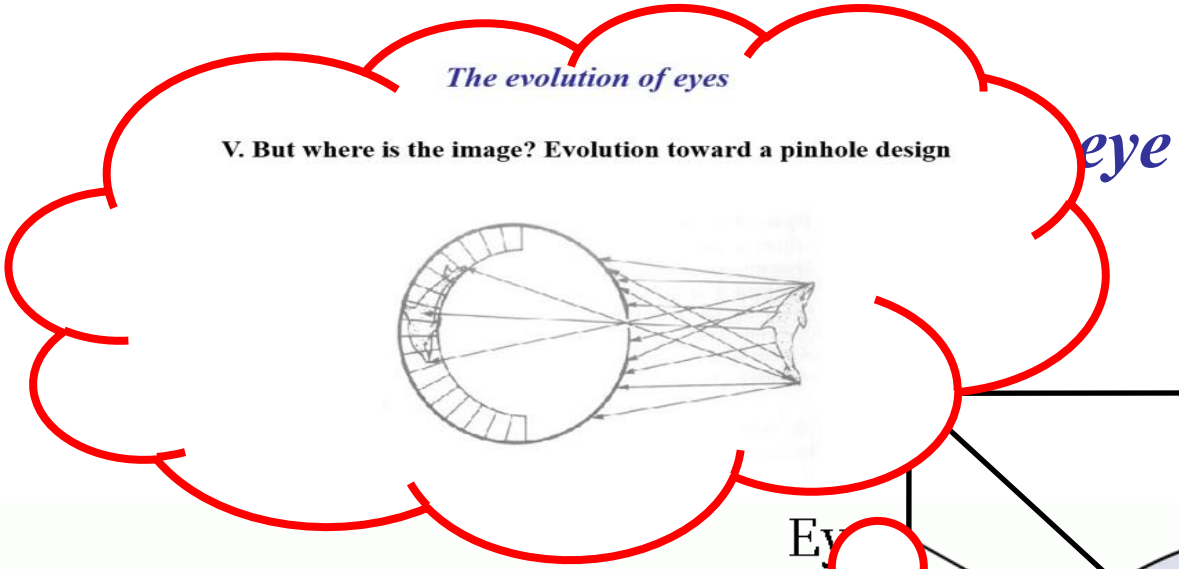




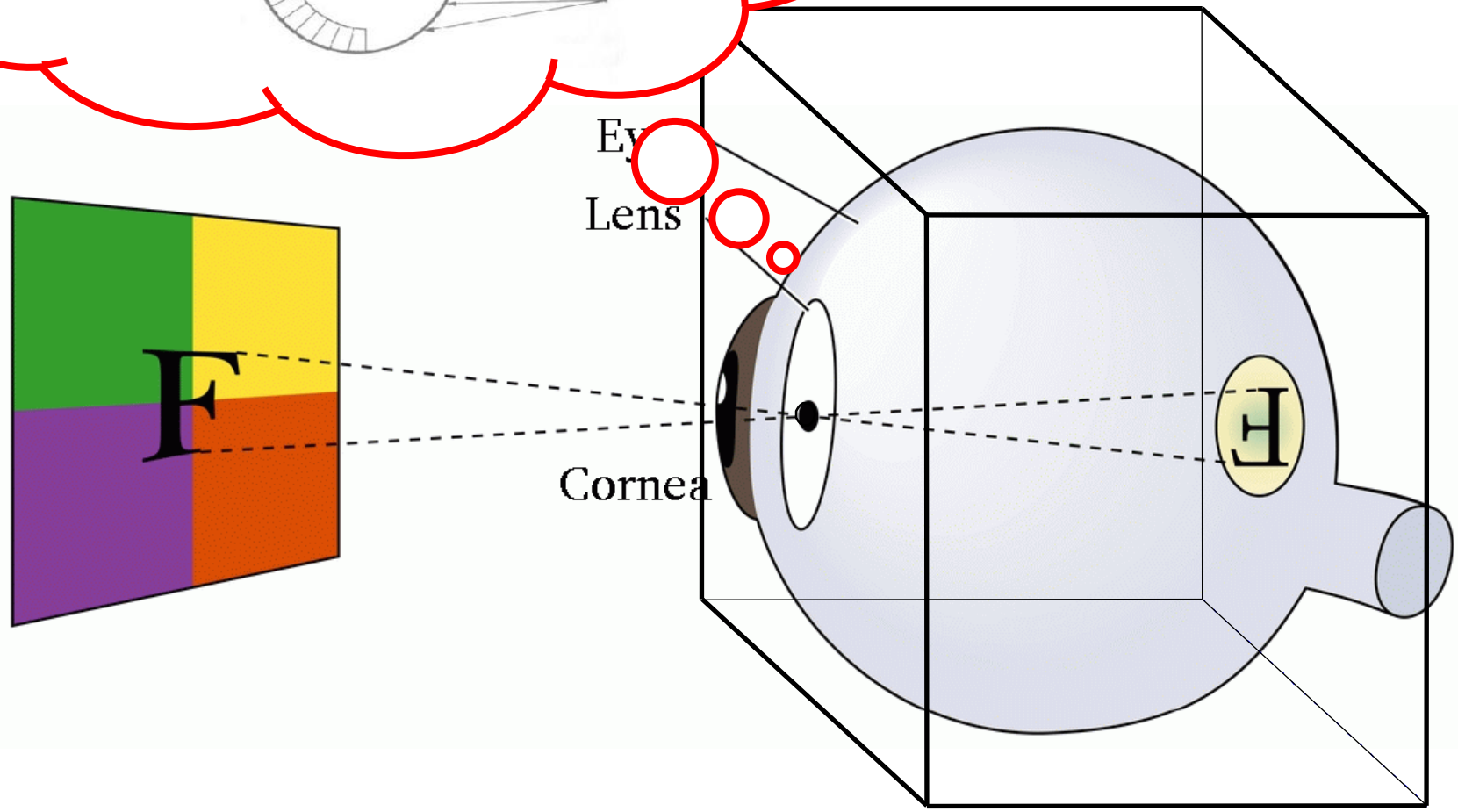
From https://microanatomy.net/Eye/retina_sclera_and_choroid.htm

The human eye





Pinhole camera model

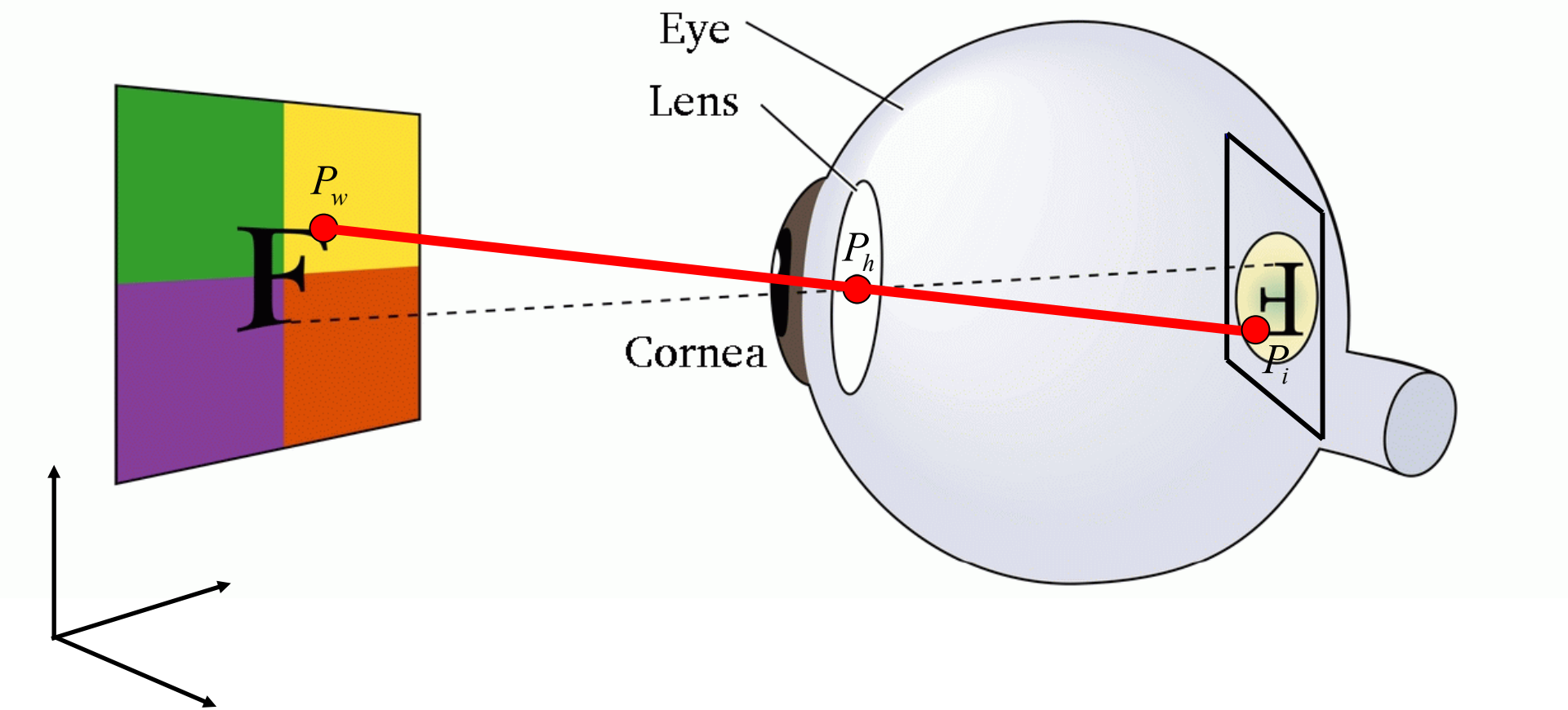


The human eye

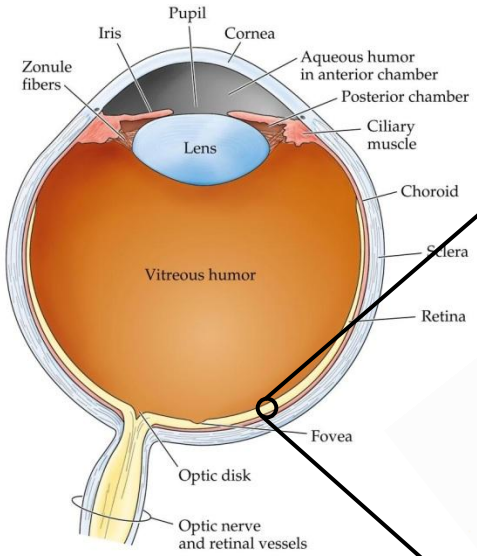
***Camera obscura (dark chamber)* – first cameras were indeed pinhole**

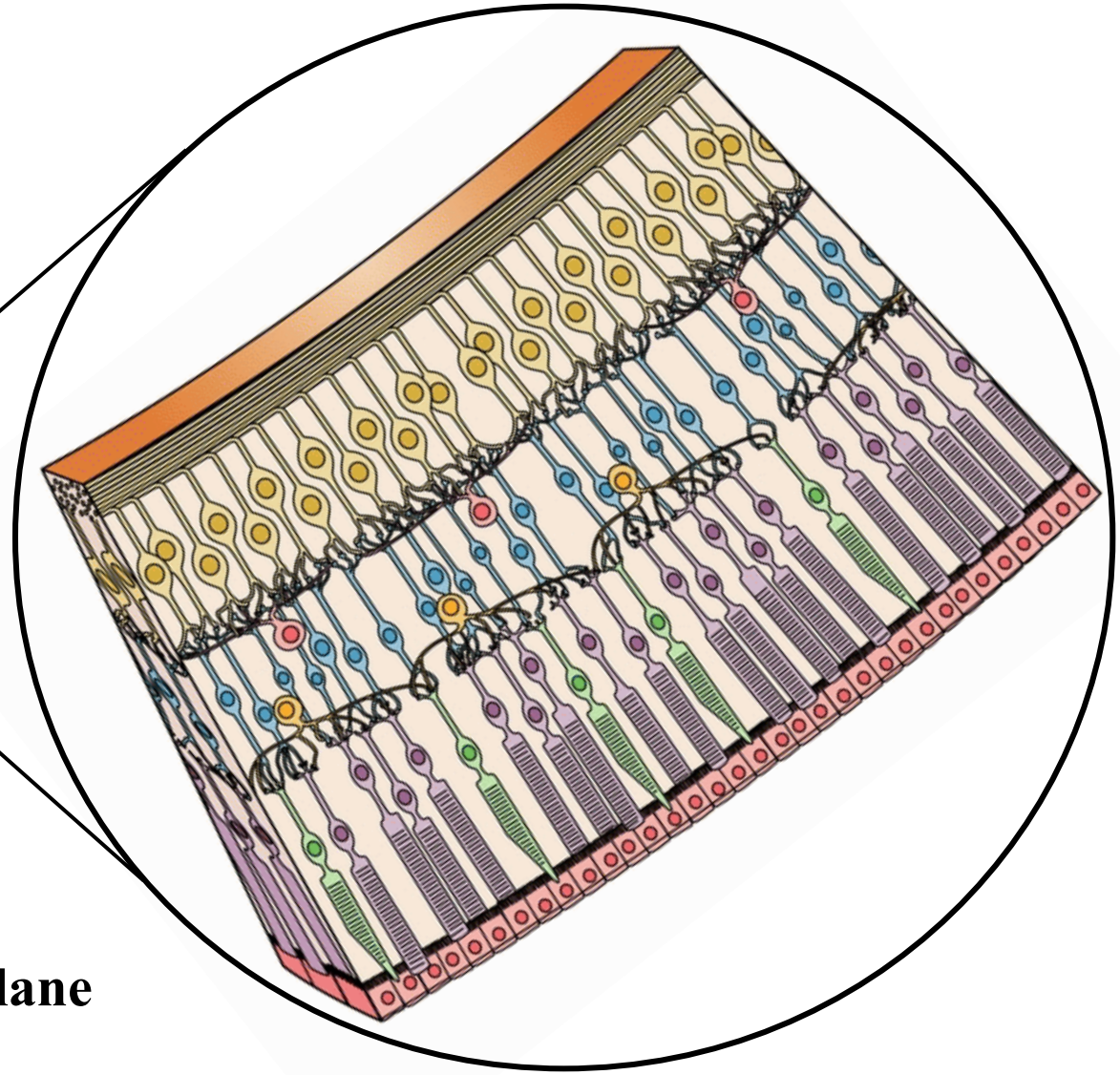
The human eye

Pinhole camera model



The human eye



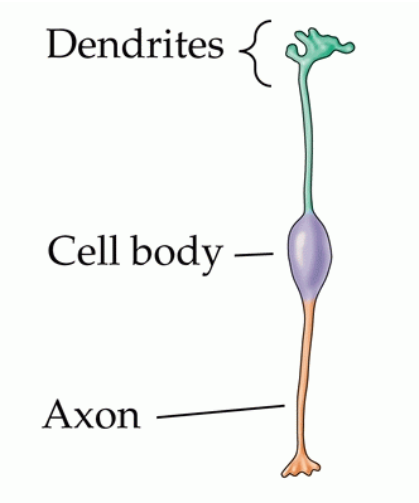


The retina:

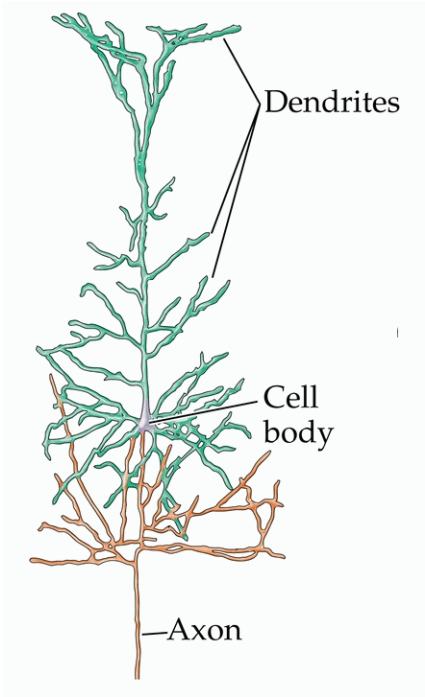
The biological image plane

The human eye

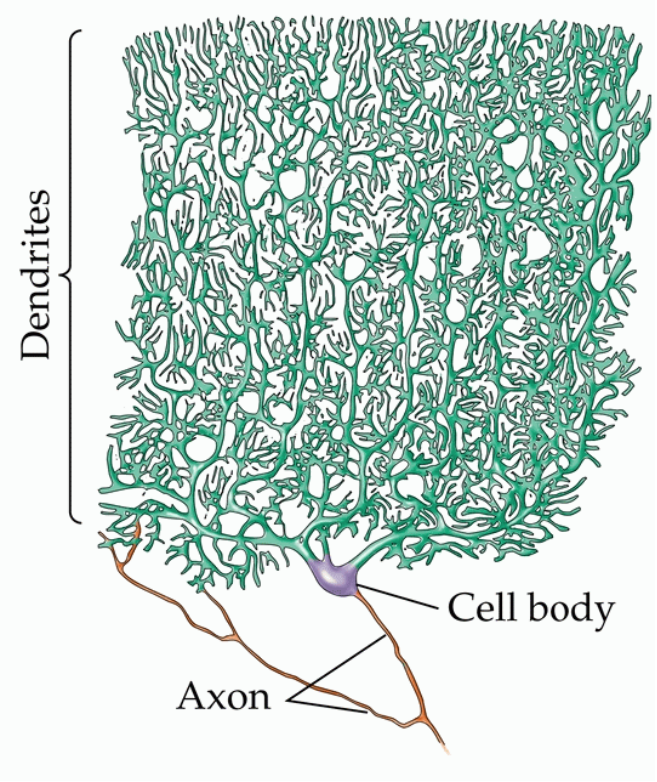
Aside – Neural signaling and communication



Retinal bipolar cell



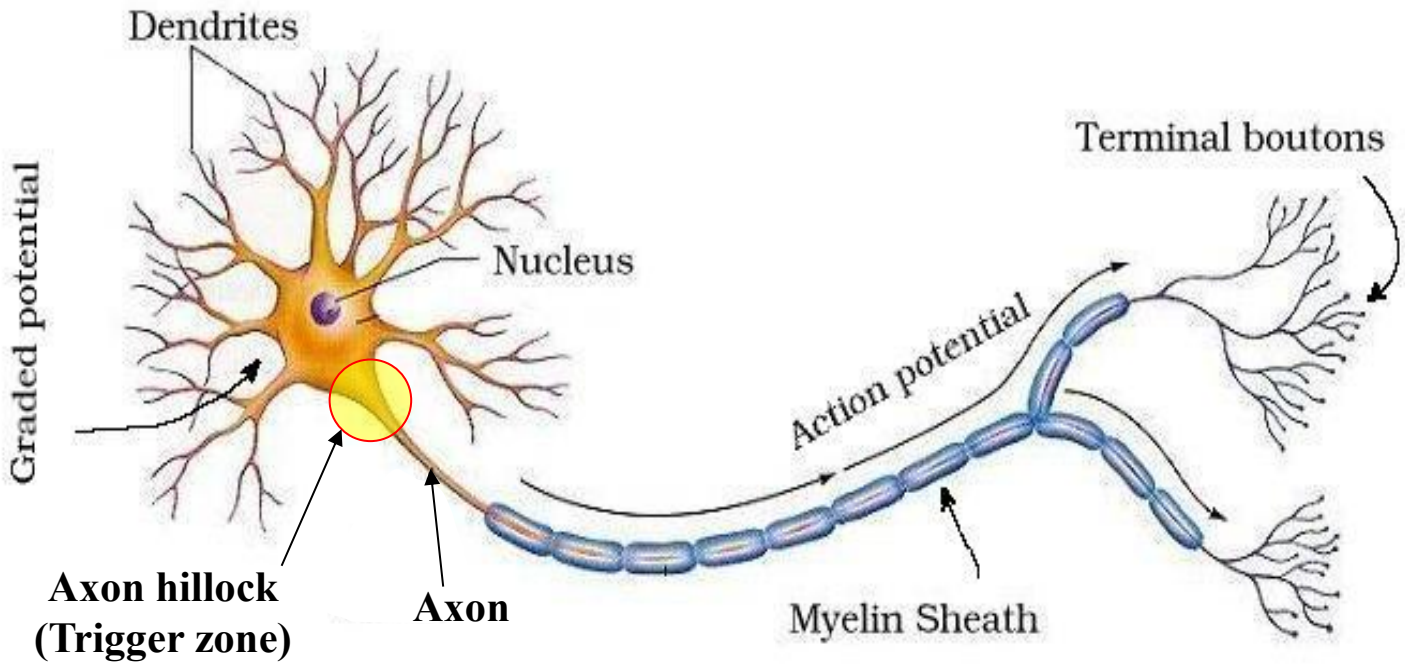
Cerebellar pyramidal cell



Cerebellar Purkinje cell

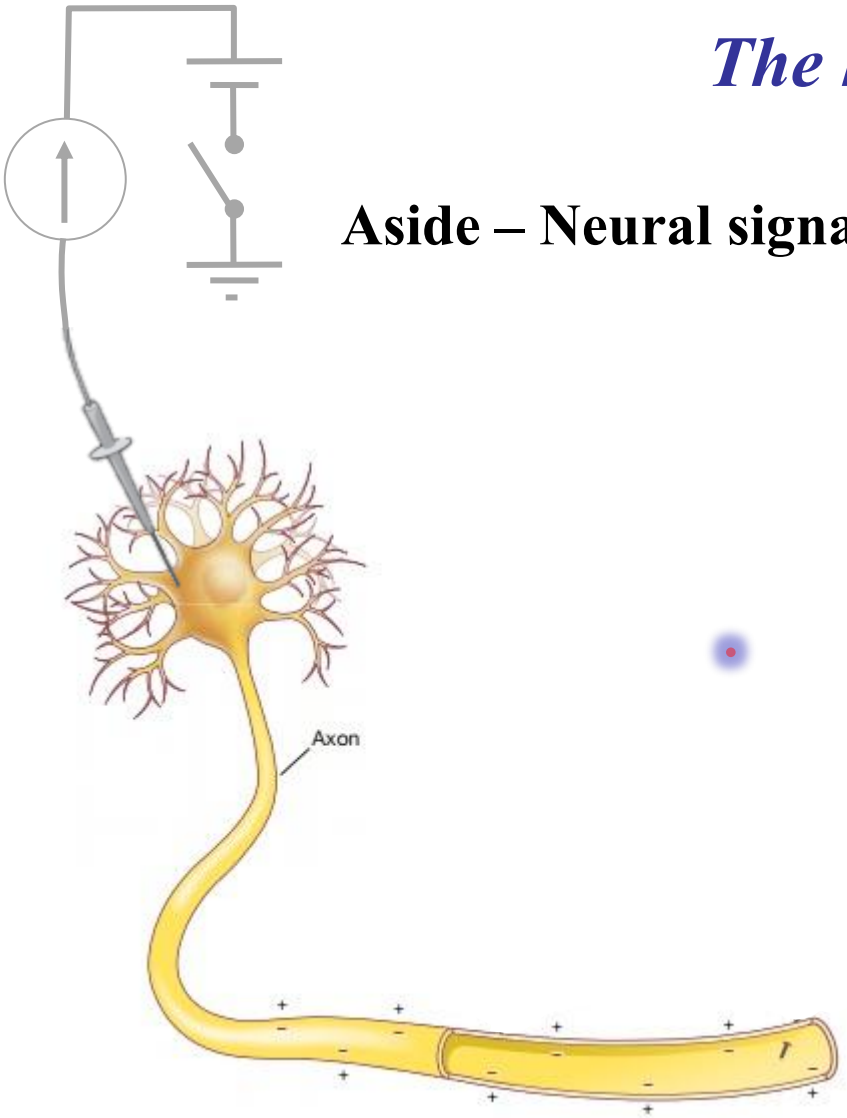
The human eye

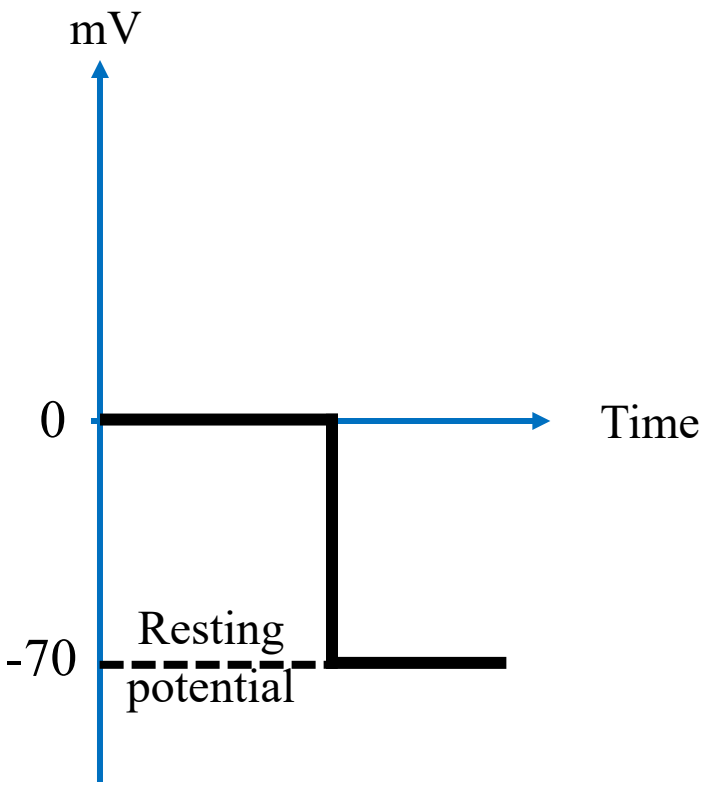
Aside – Neural signaling and communication



The human eye

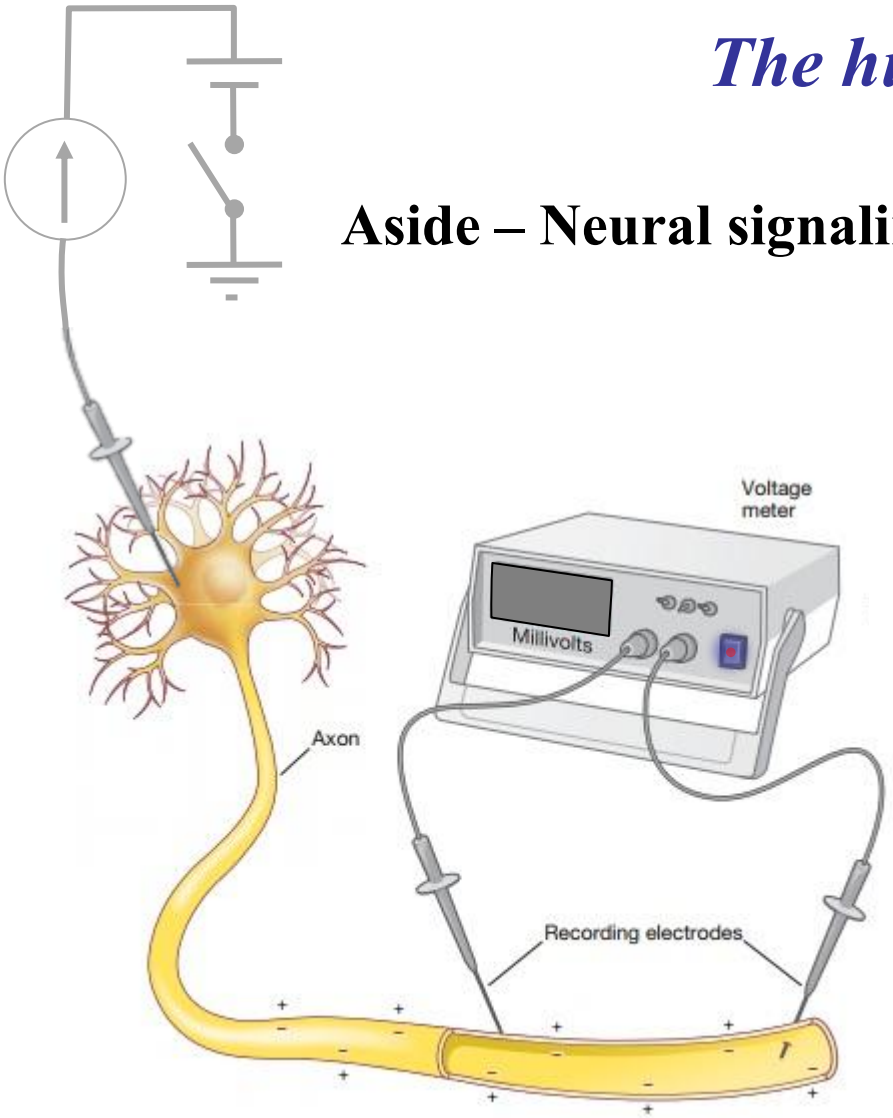
Aside – Neural signaling and communication

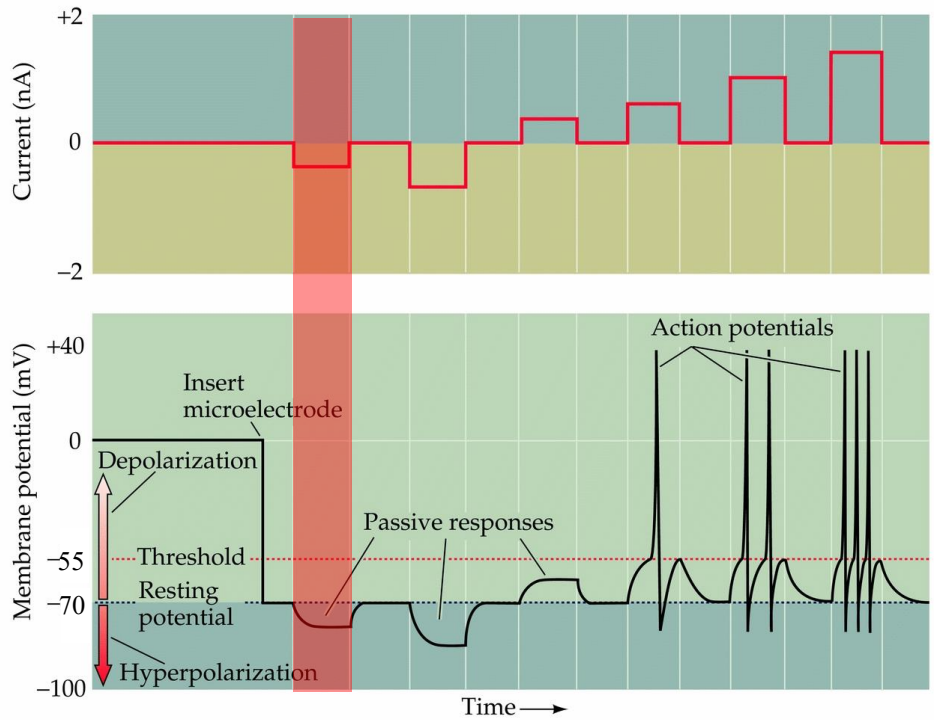




The human eye

Aside – Neural signaling and communication





The human eye

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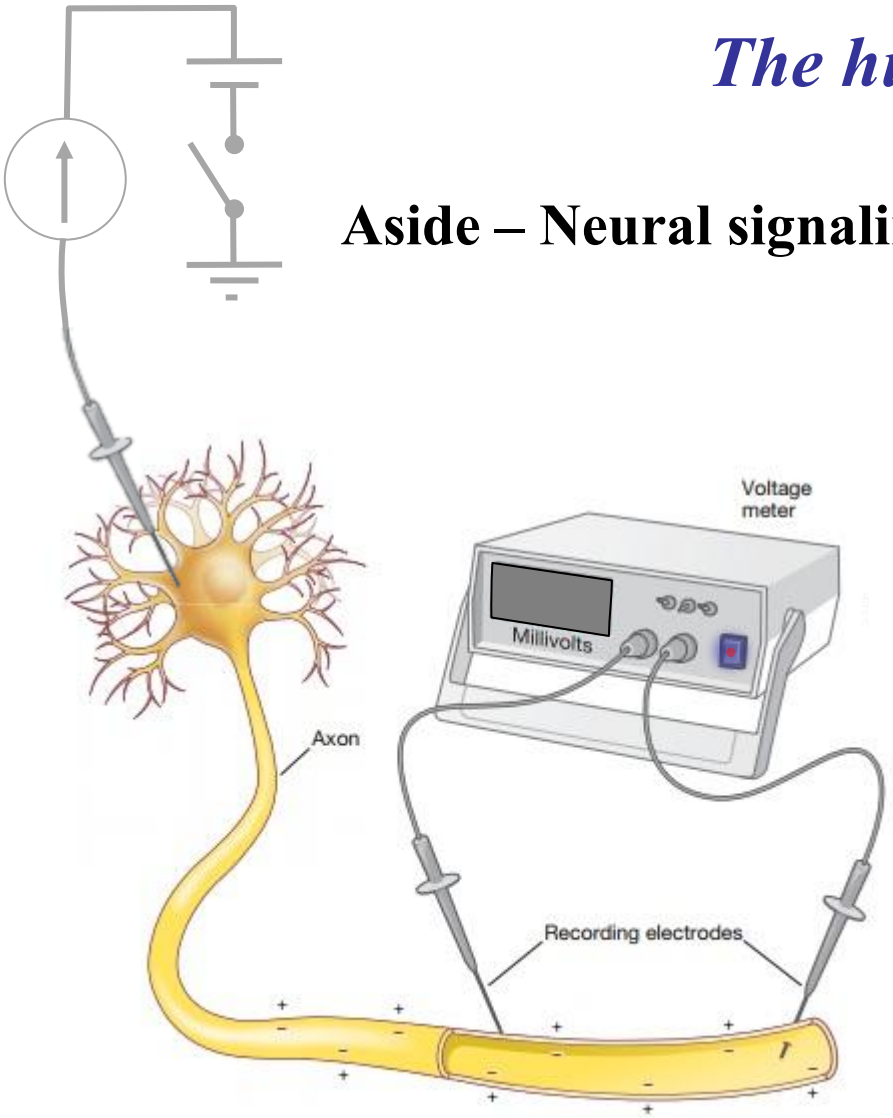

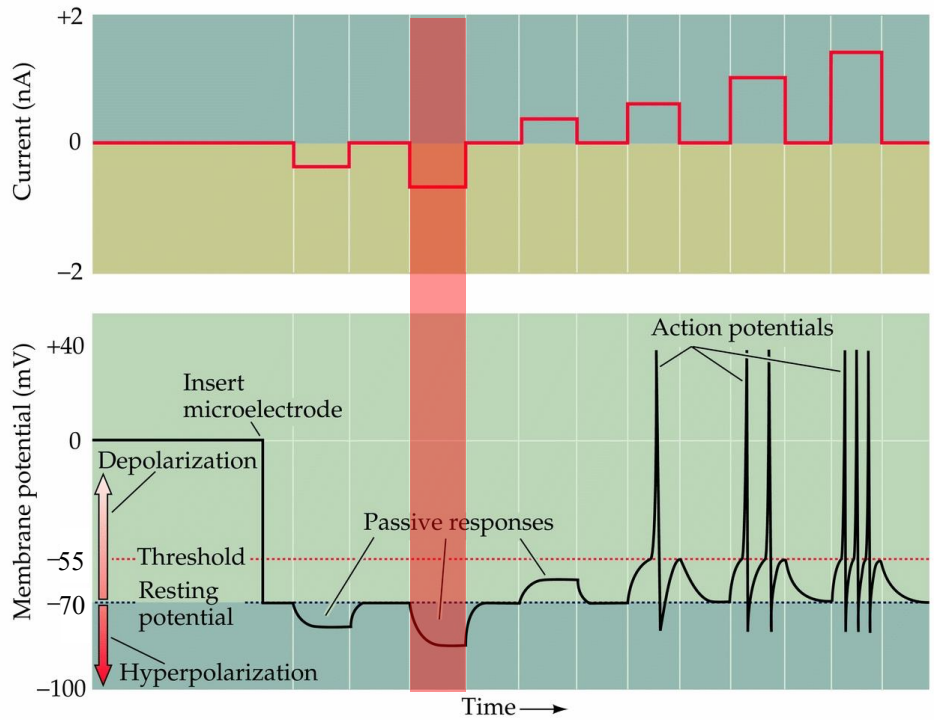

The human eye

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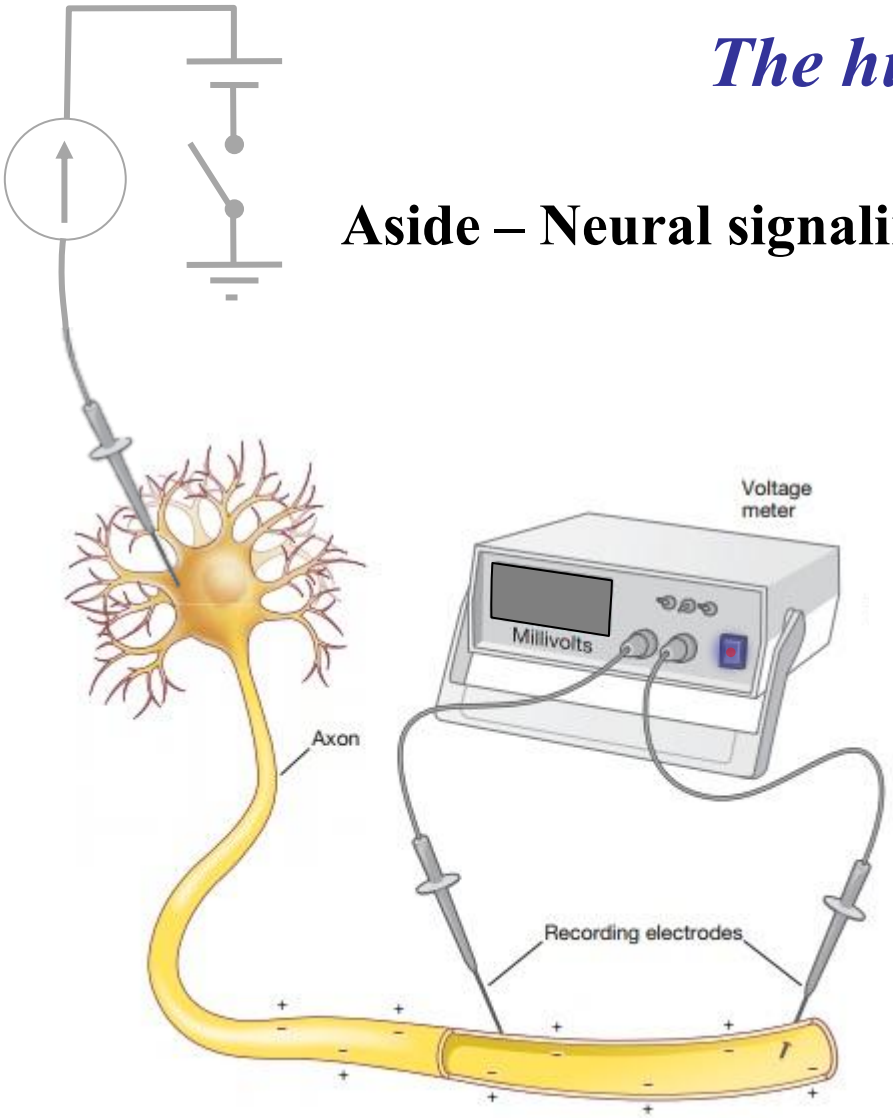
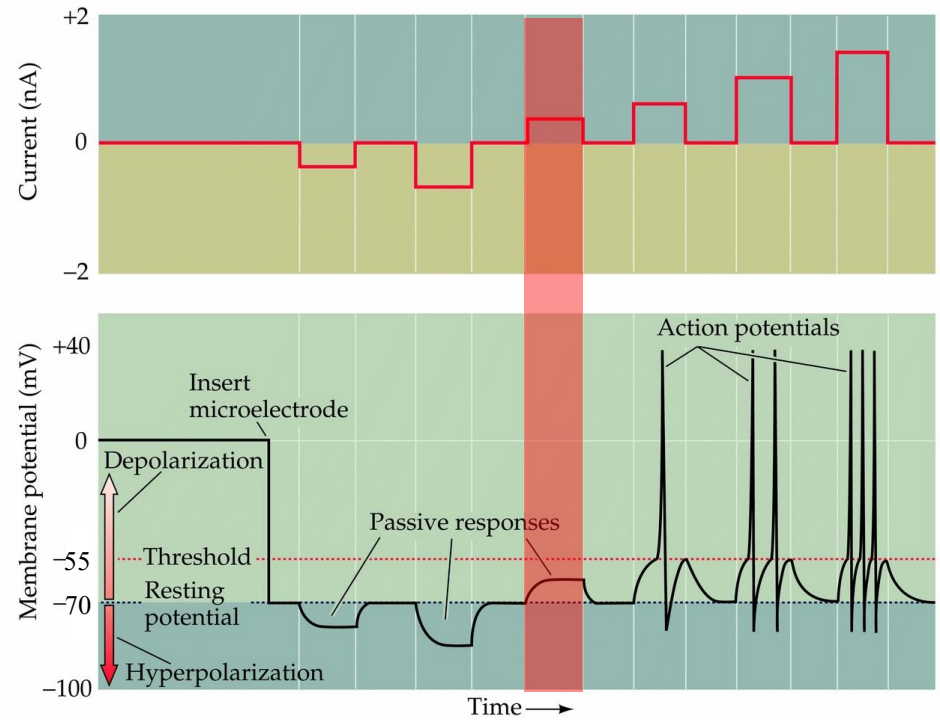
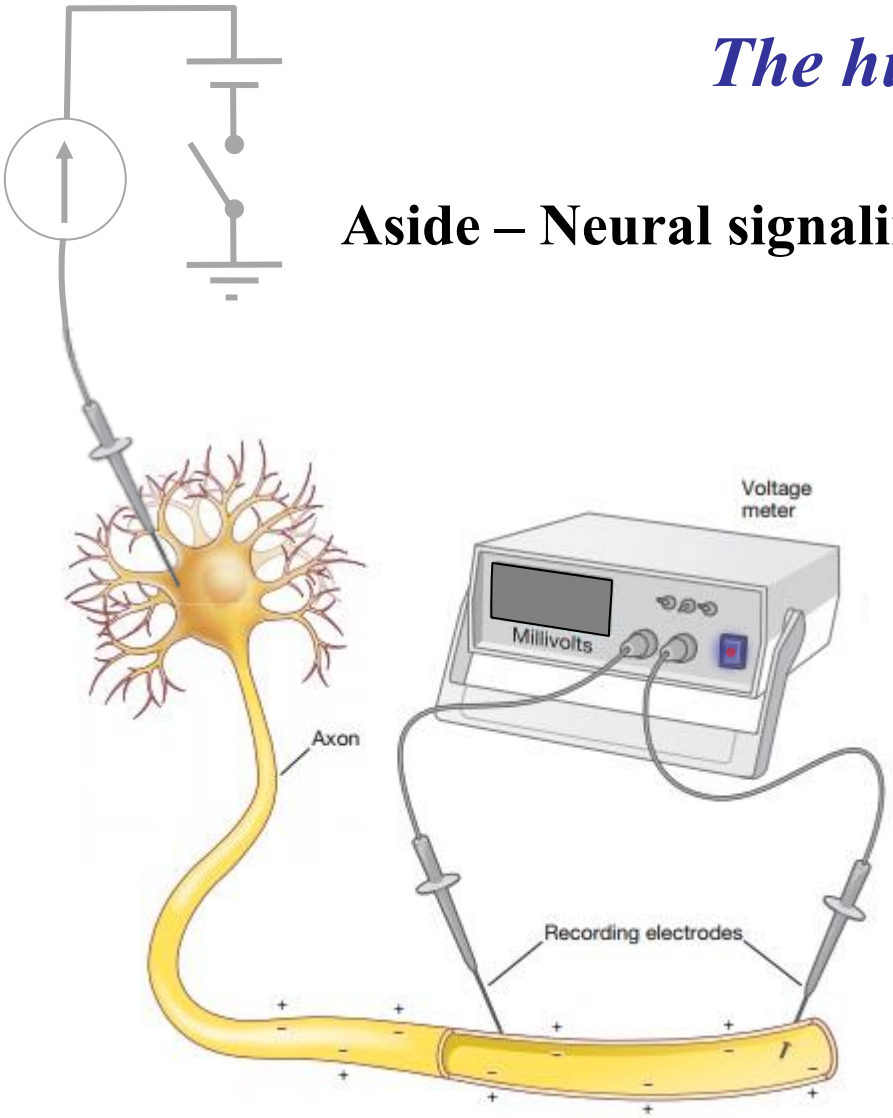
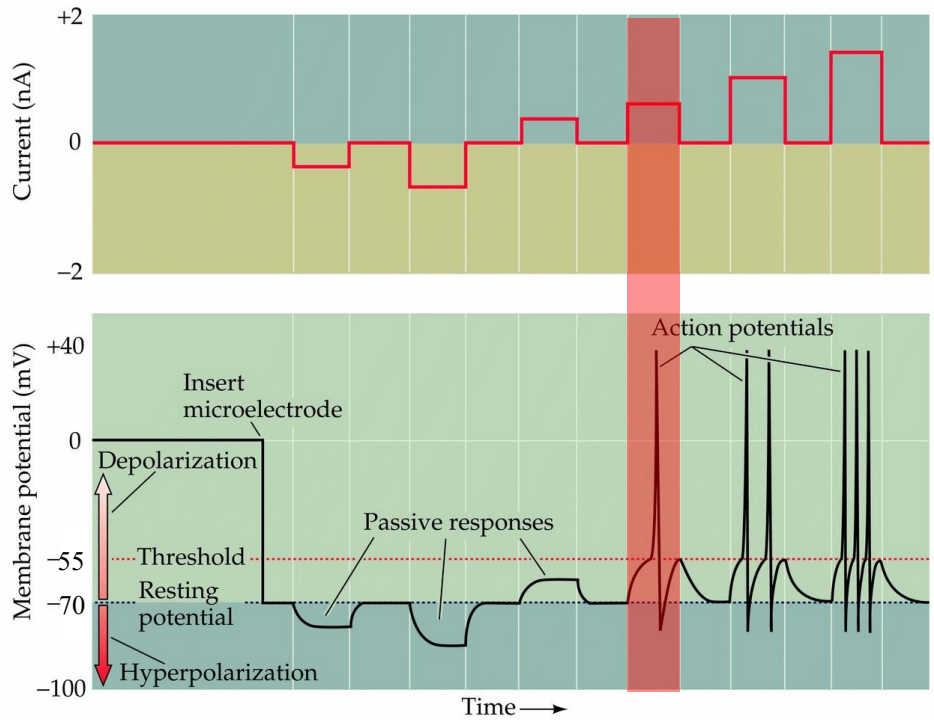




The human eye

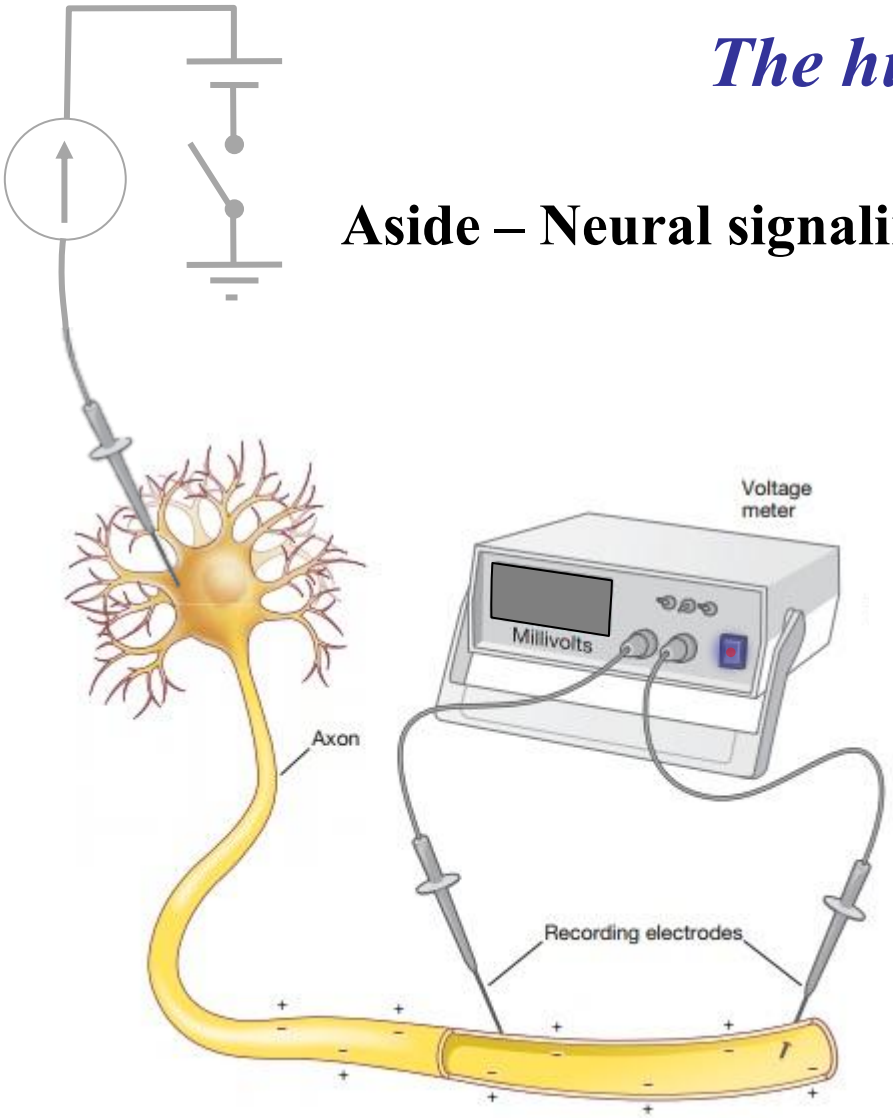
Aside – Neural signaling and communication

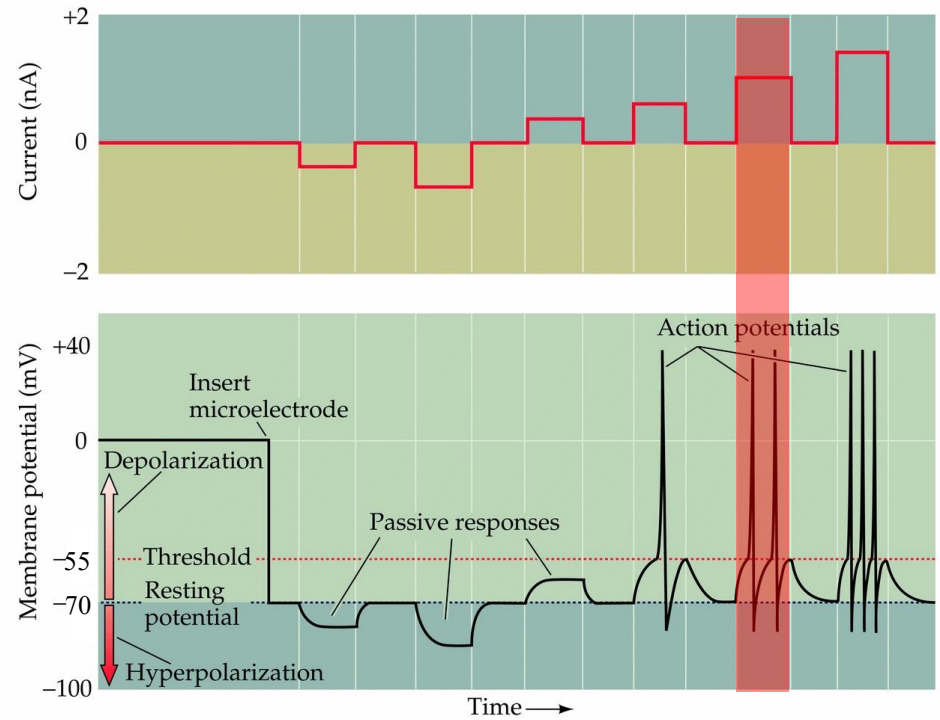




The human eye

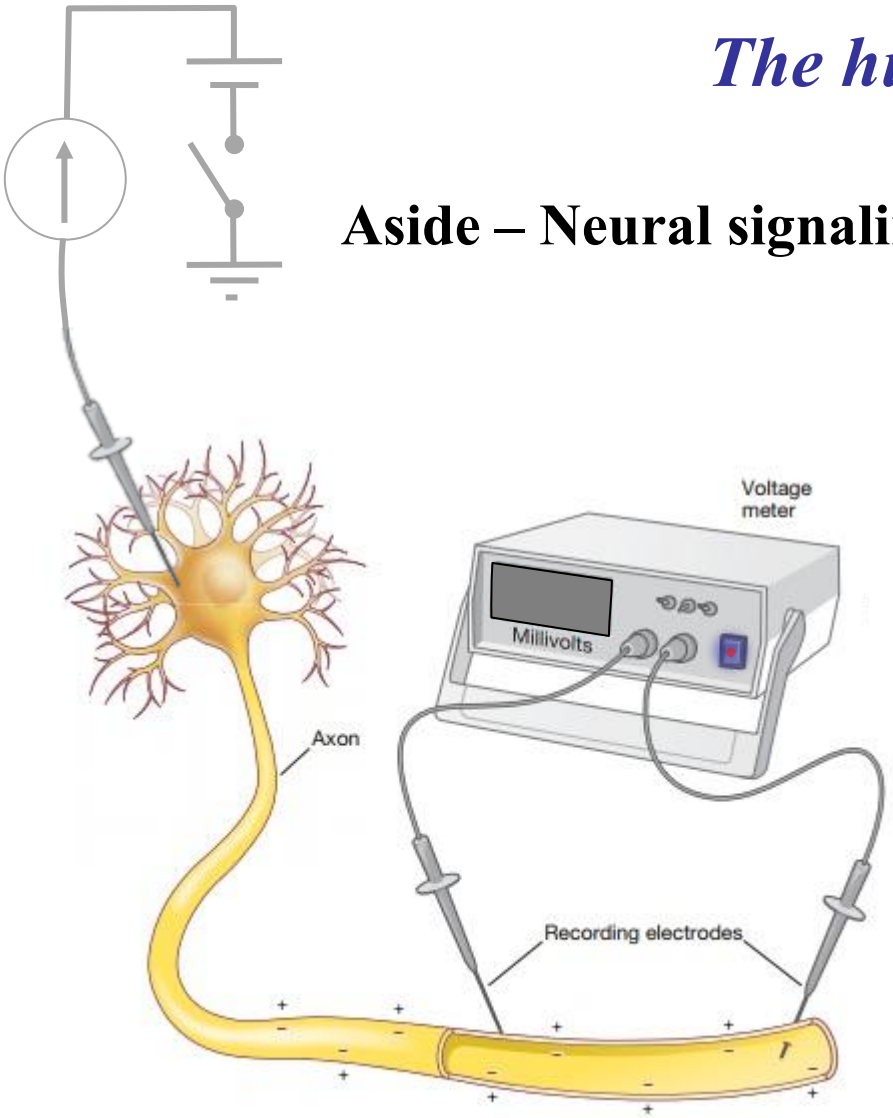
Aside – Neural signaling and communication

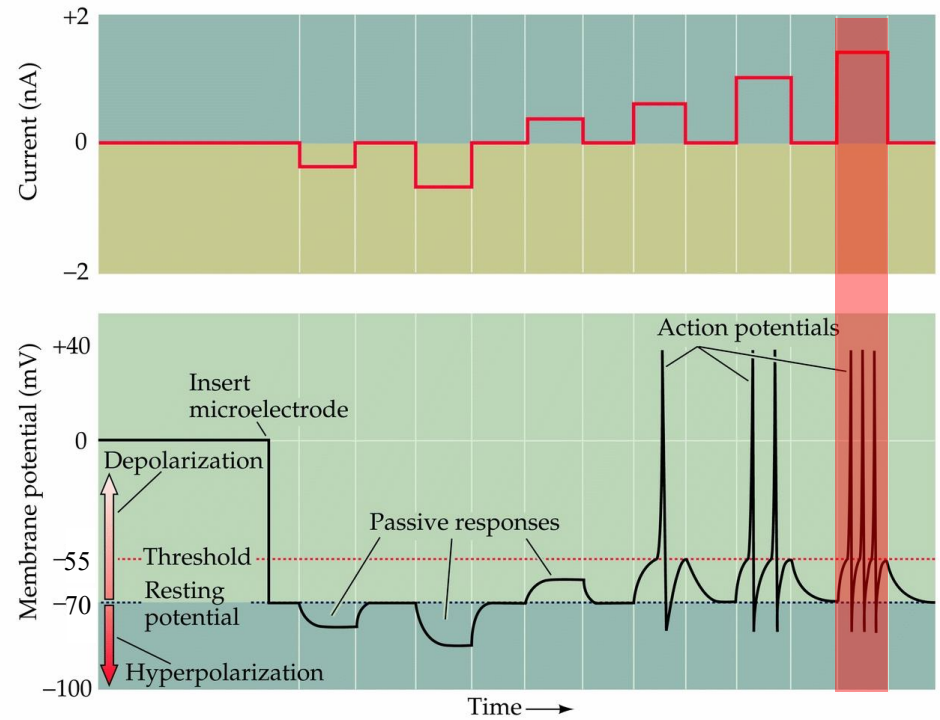




The human eye

Aside – Neural signaling and communication

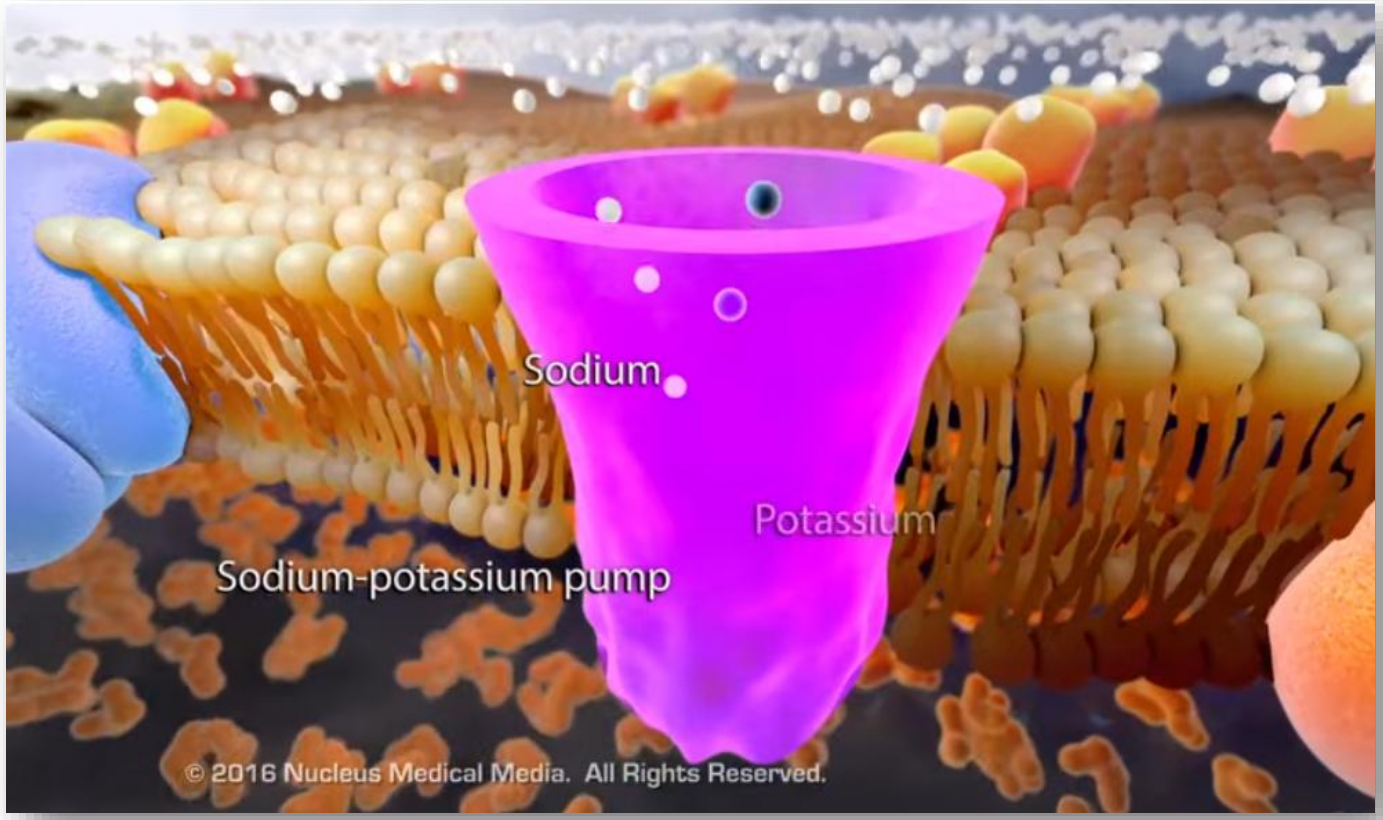




The human eye

Aside – Neural signaling and communication

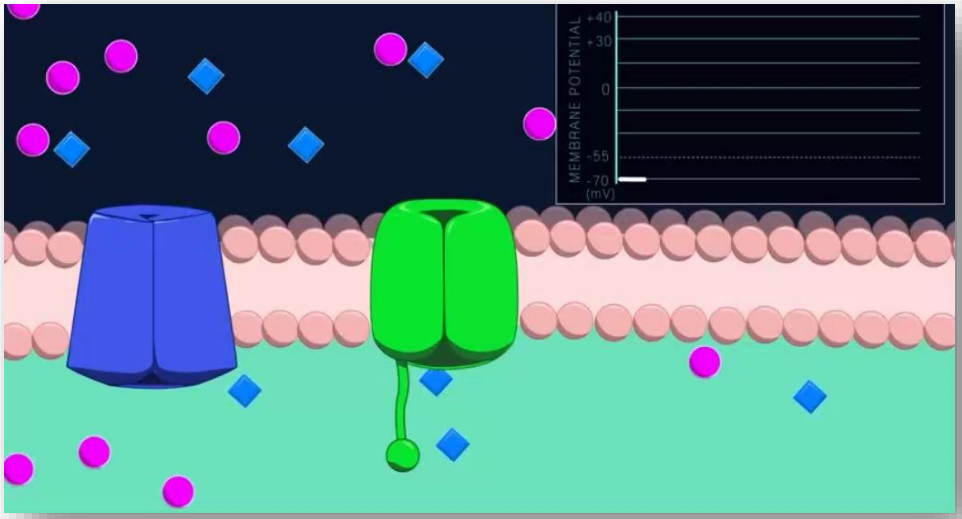
Mechanisms I: Sodium-Potassium pump

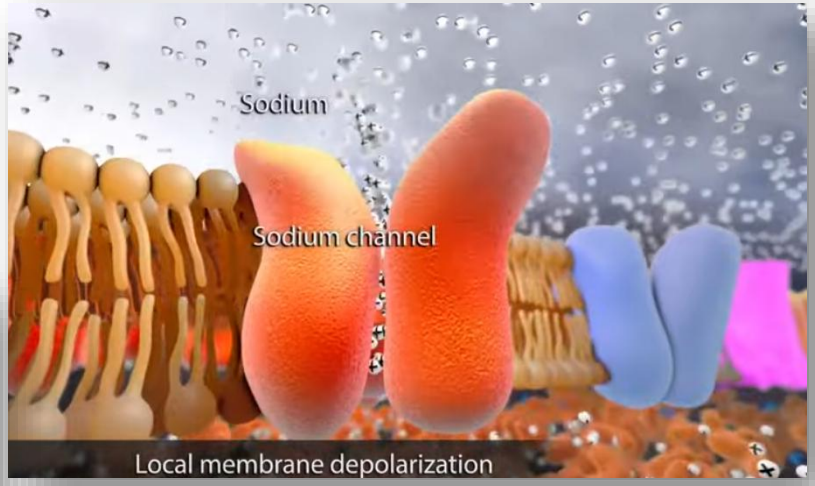


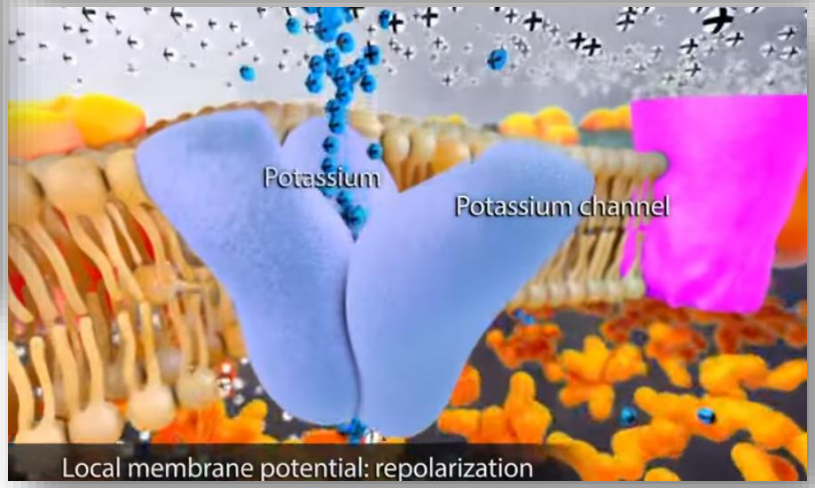
The human eye

Aside – Neural signaling and communication

Mechanisms II: Voltage-gated channels



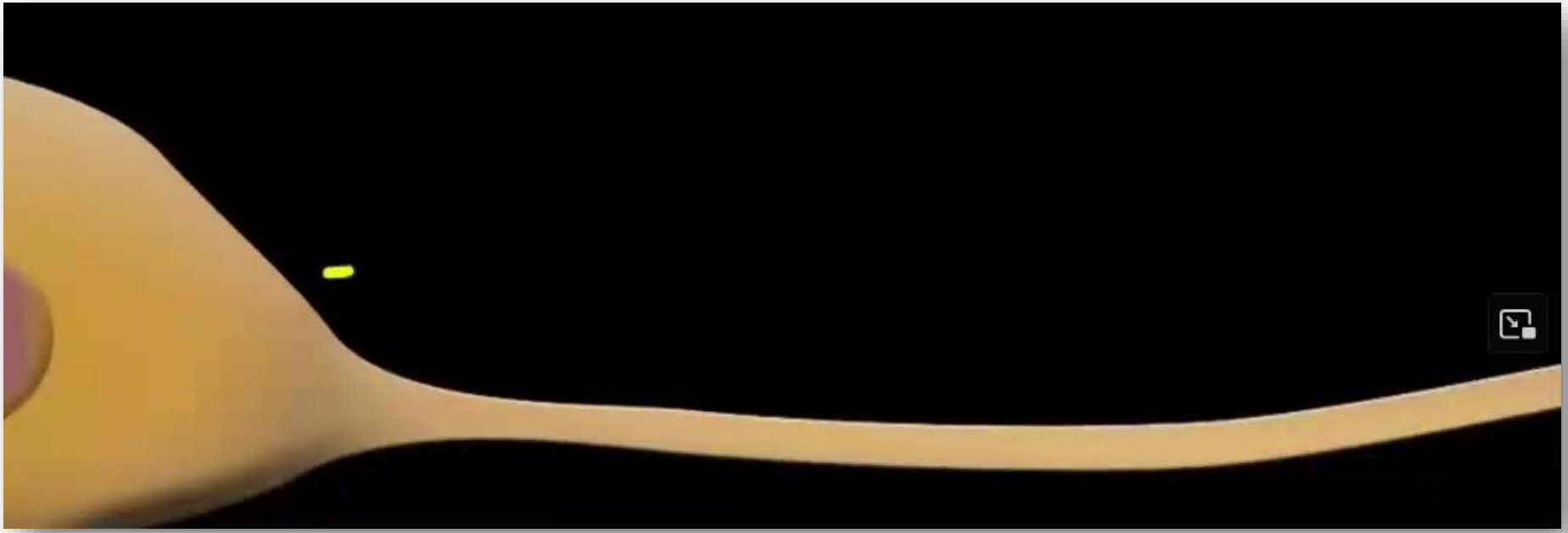




The human eye

Aside – Neural signaling and communication

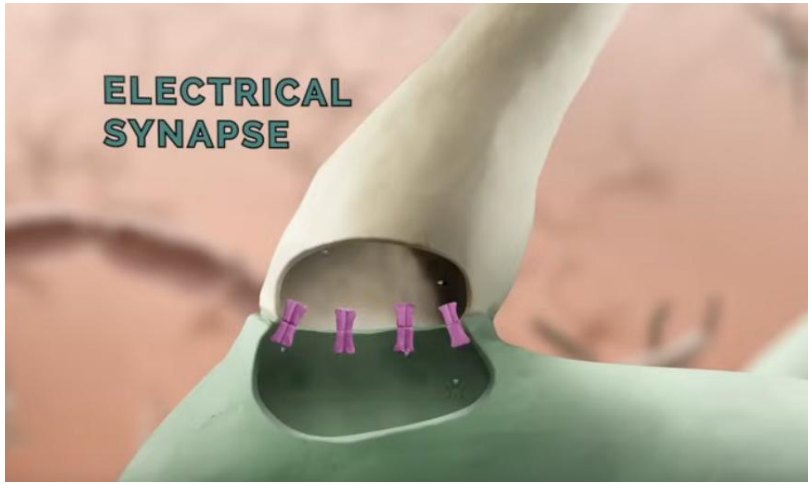
Propagation of action potential along the axon:

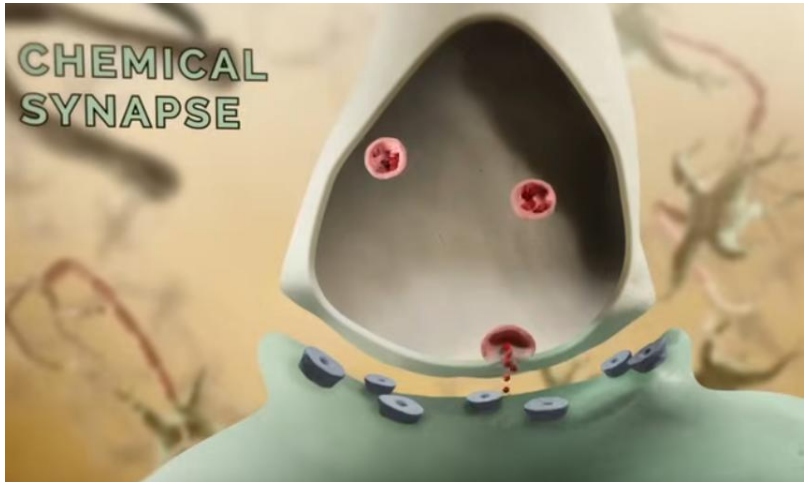


The human eye

Aside – Neural signaling and communication

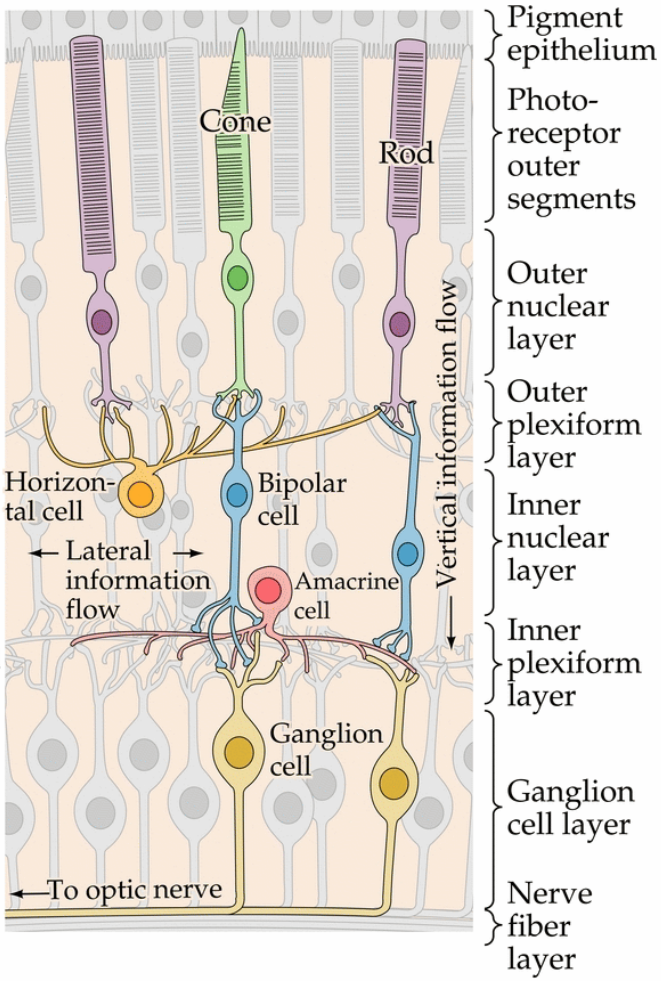
Synapses: where it all begins, and ends





The human eye

The retina



Variety of neuron types:

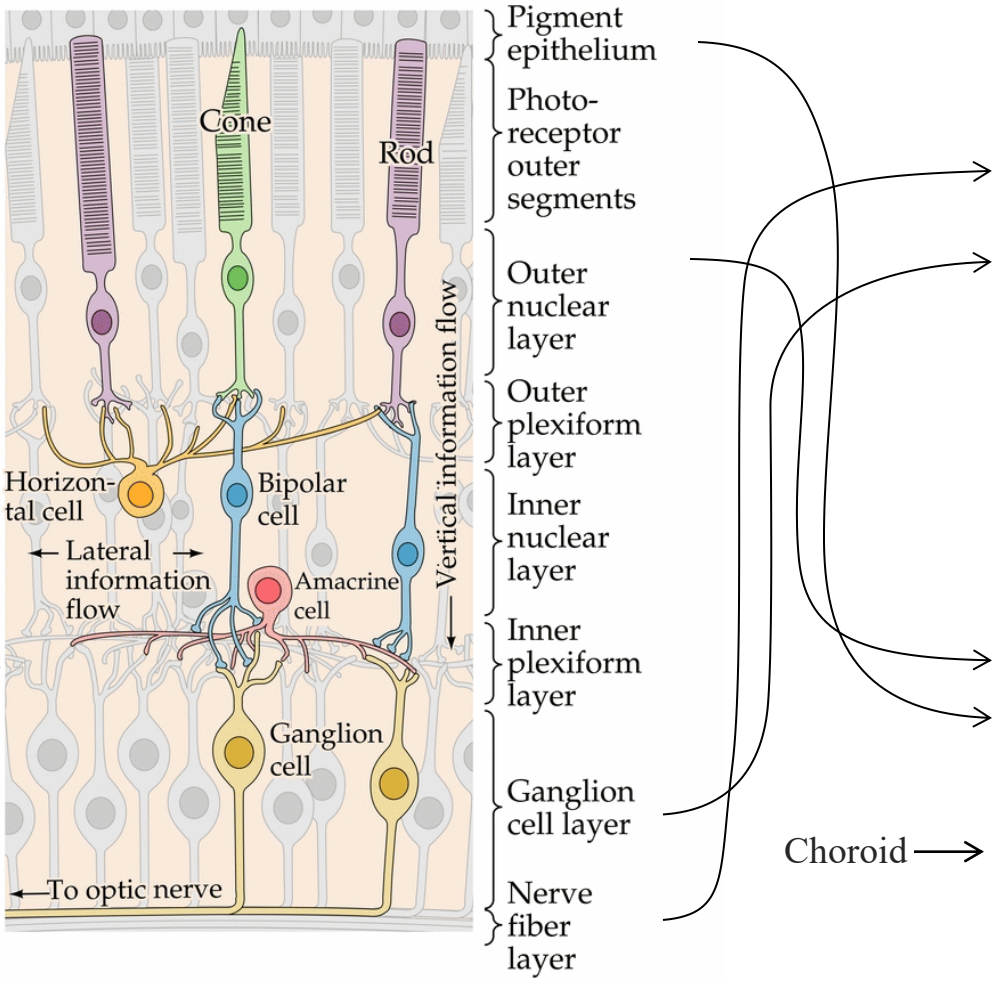
- **Photoreceptors: ~100 million (95% rods,5% cones)**
- **Bipolar cells**
- **Ganglion cells**
- **Amacrine cells**
- **Horizontal cells**

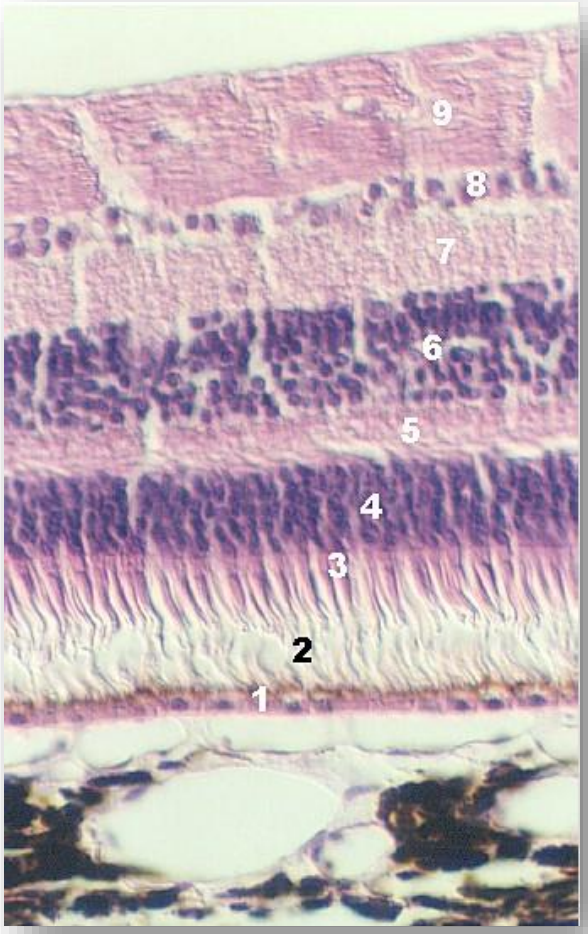
Optic nerve:

- **Made of axons of secondary (ganglion) neurons**
- **~1 million axons**

The human eye

The retina

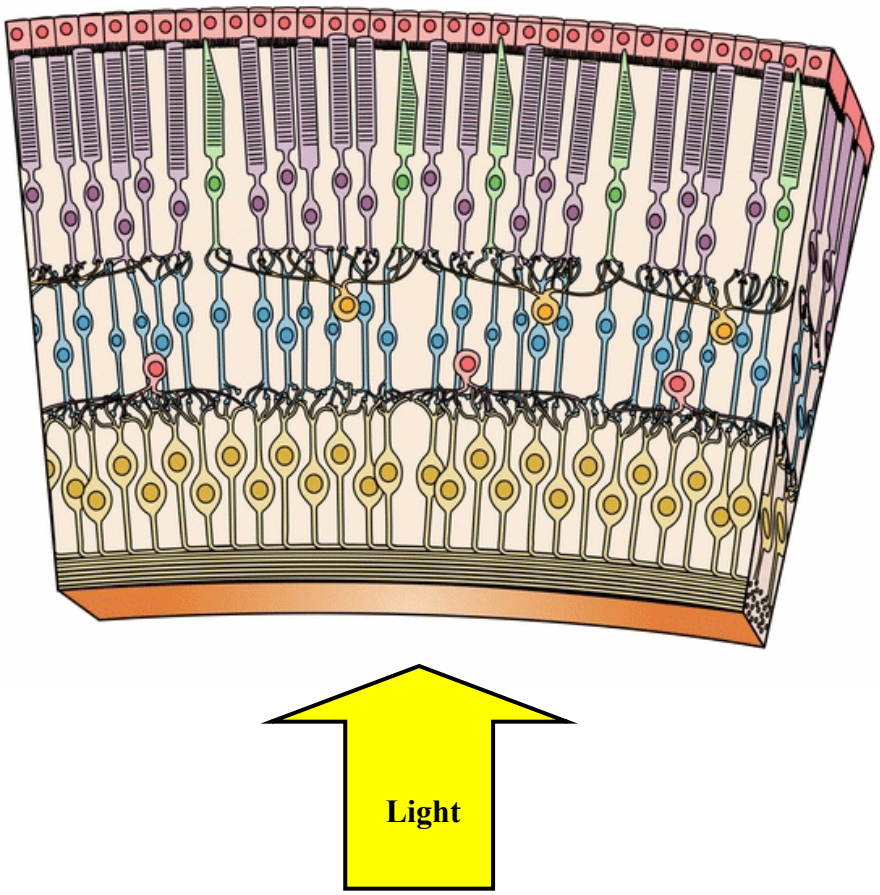




From https://microanatomy.net/Eye/retina_sclera_and_choroid.htm

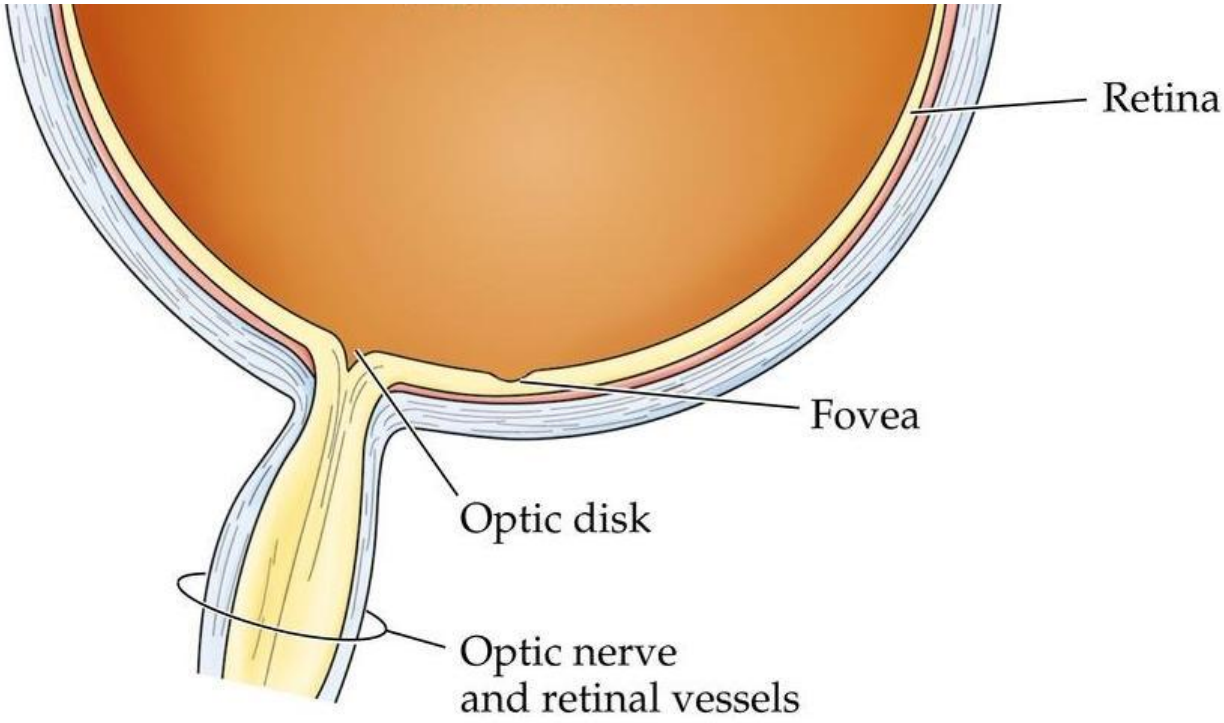
The human eye

A defective design?



The human eye

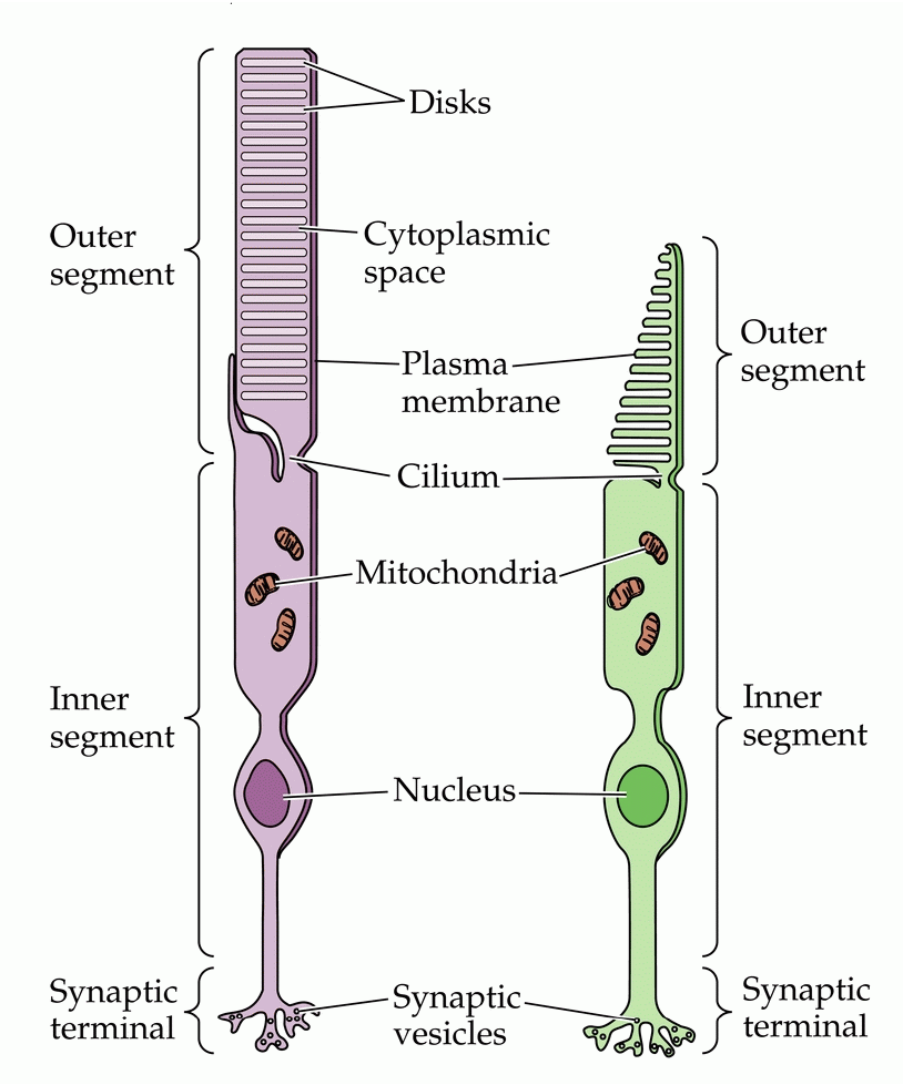
The optic disk and the blind spot



The human eye

The optic disk and the blind spot

The human eye



Rods and cones

Rods:

- **~95M in number**
- **Extremely sensitive to light**
- **Single photon response**
- **Low spatial resolution**
- **Single response profile**
- **B&W night vision (scotopic)**

Cones:

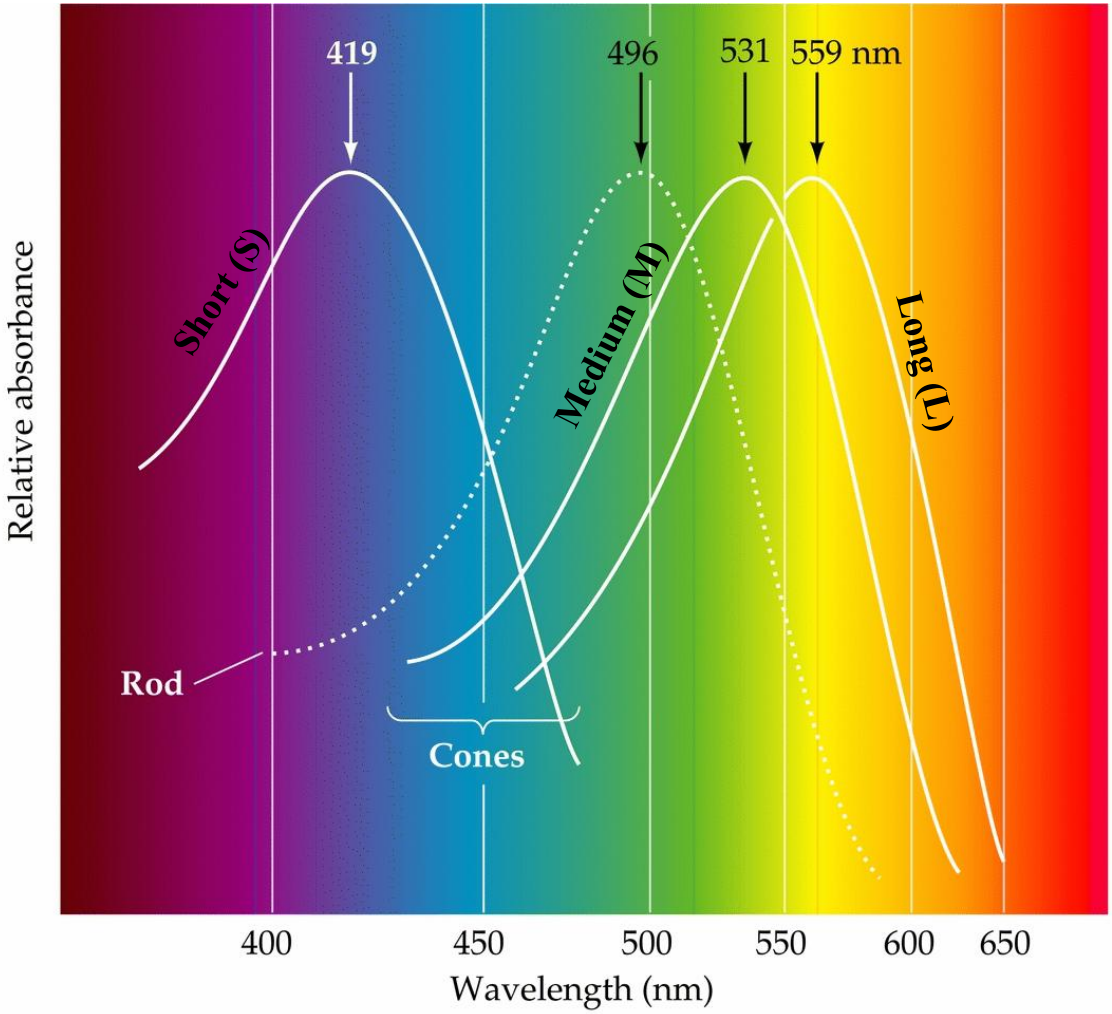
- **~5M in number**
- **Relatively insensitive to light**
- **100 photos for response comparable with rods**
- **High spatial resolution**
- **Different (3 types of) response profiles**
- **Color daylight vision (photopic)**

The human eye

Rods and cones

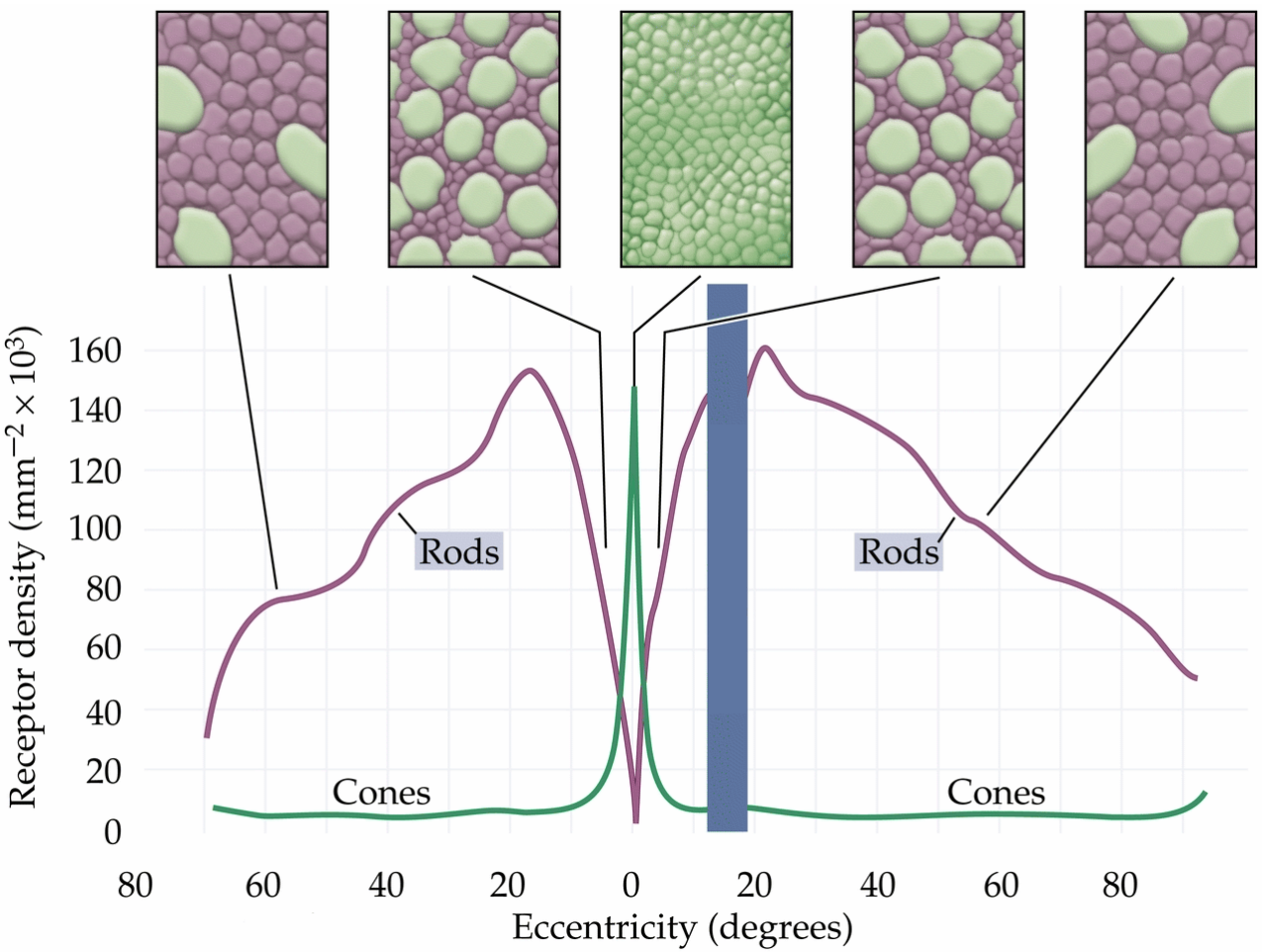
The human eye

Rods and cones



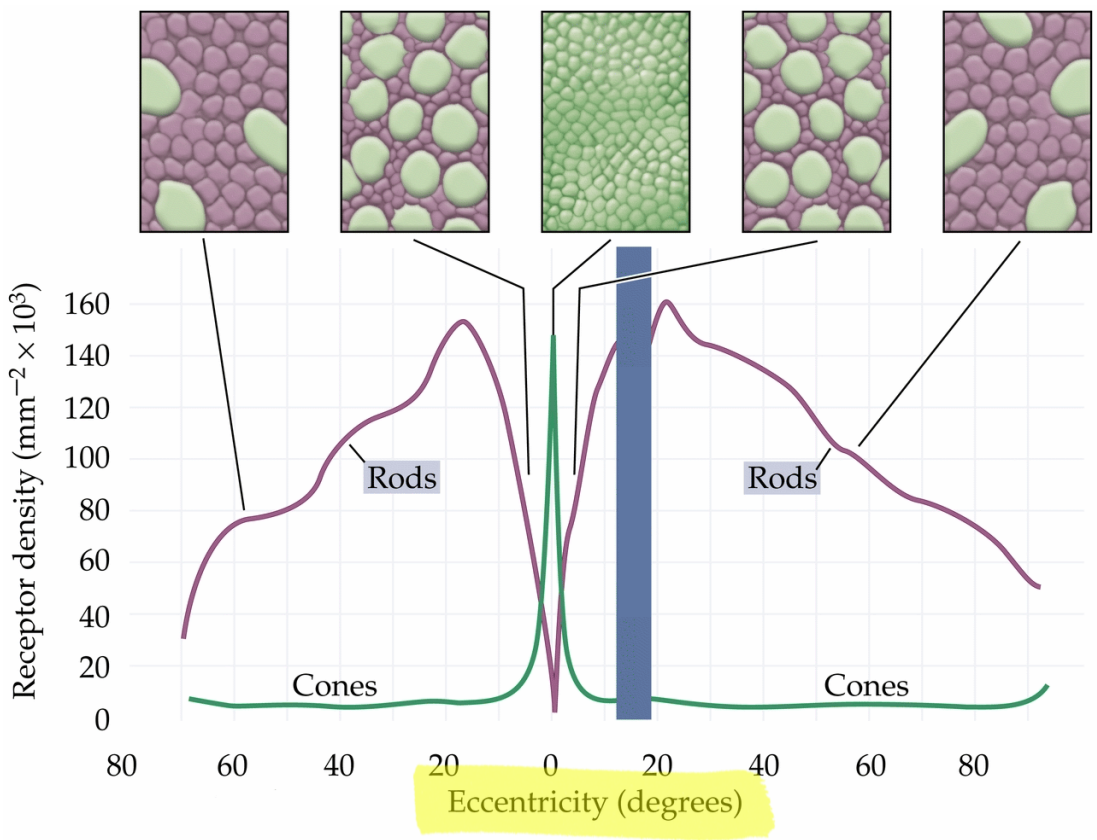
The human eye

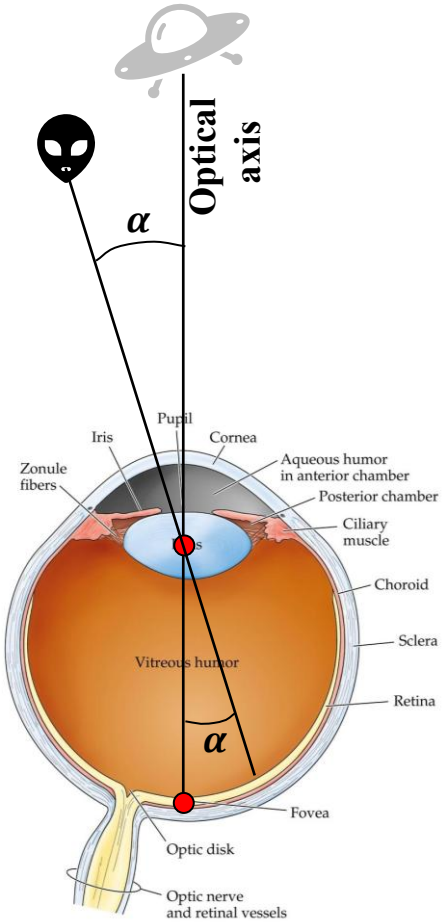
Rods and cones: distribution across the retina



The human eye

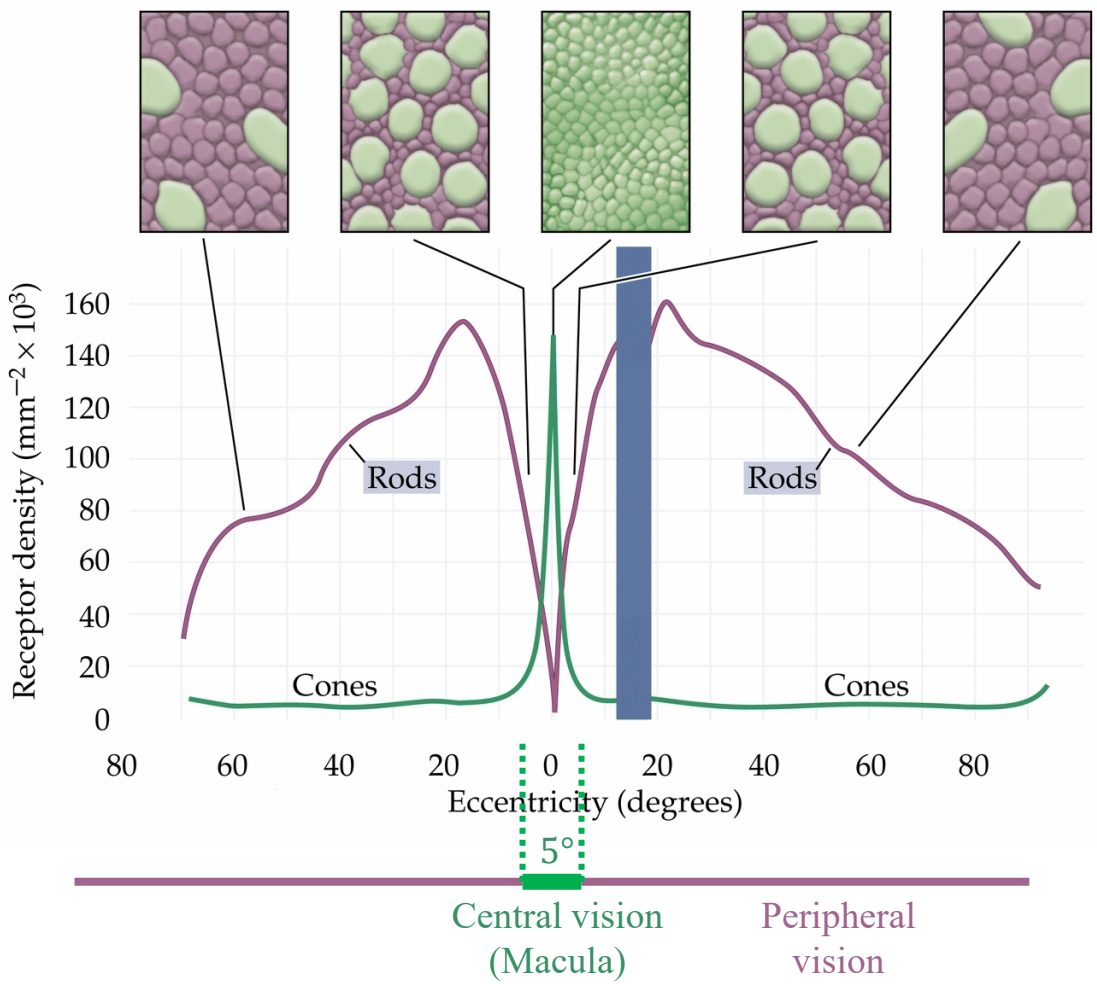
Rods and cones: distribution across the retina

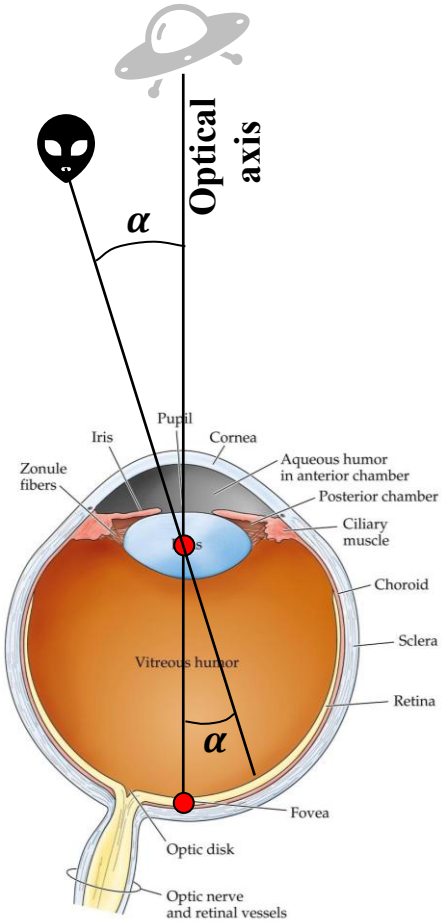




The human eye

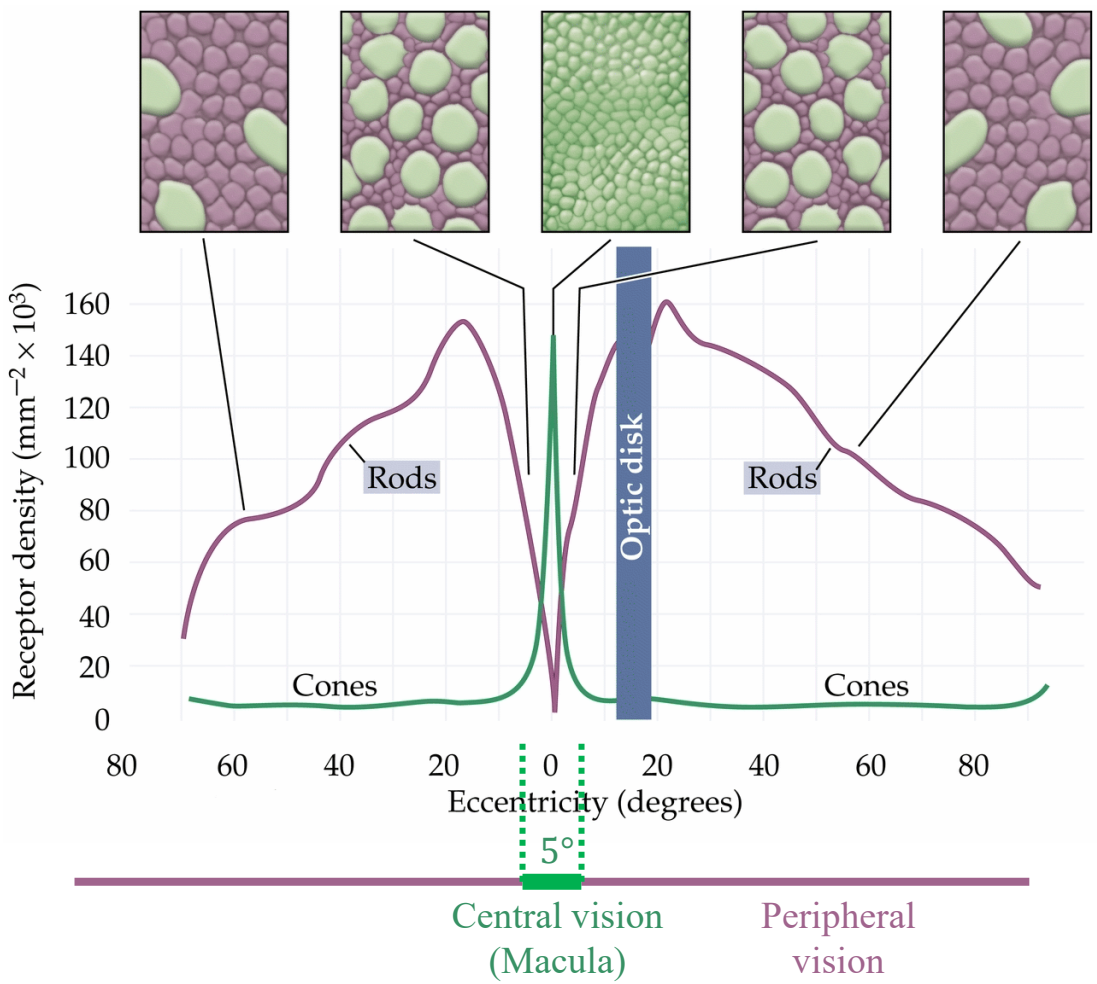
Rods and cones: distribution across the retina

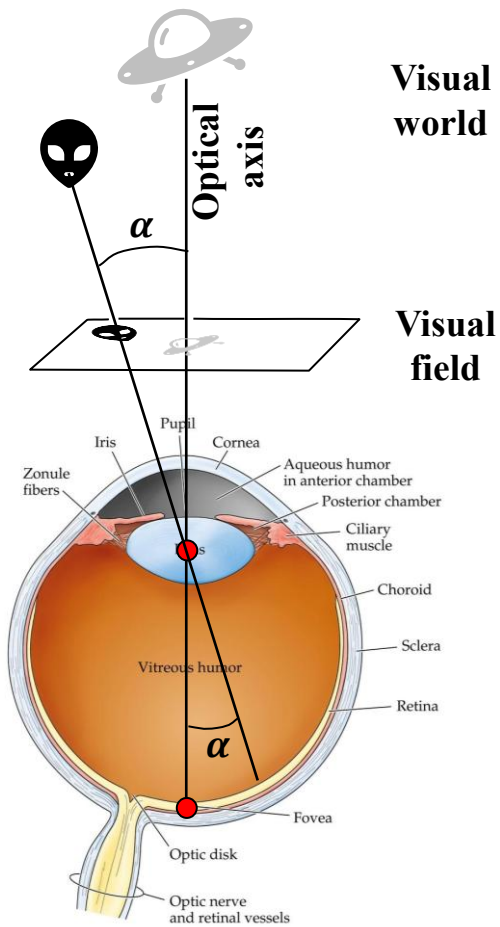




The human eye

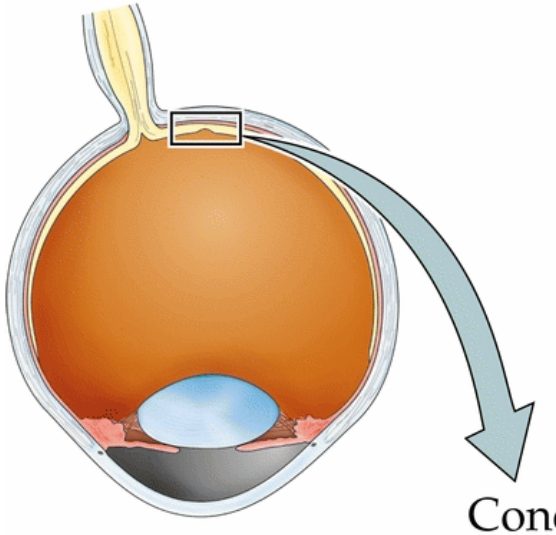
Rods and cones: distribution across the retina



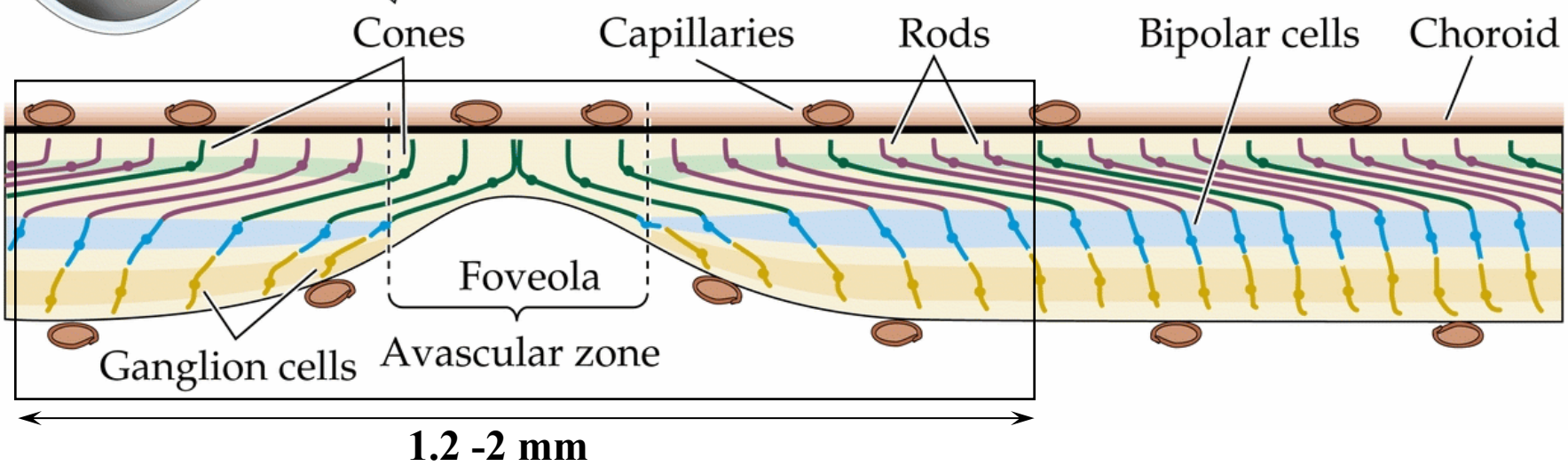


The human eye

The fovea ("pit")

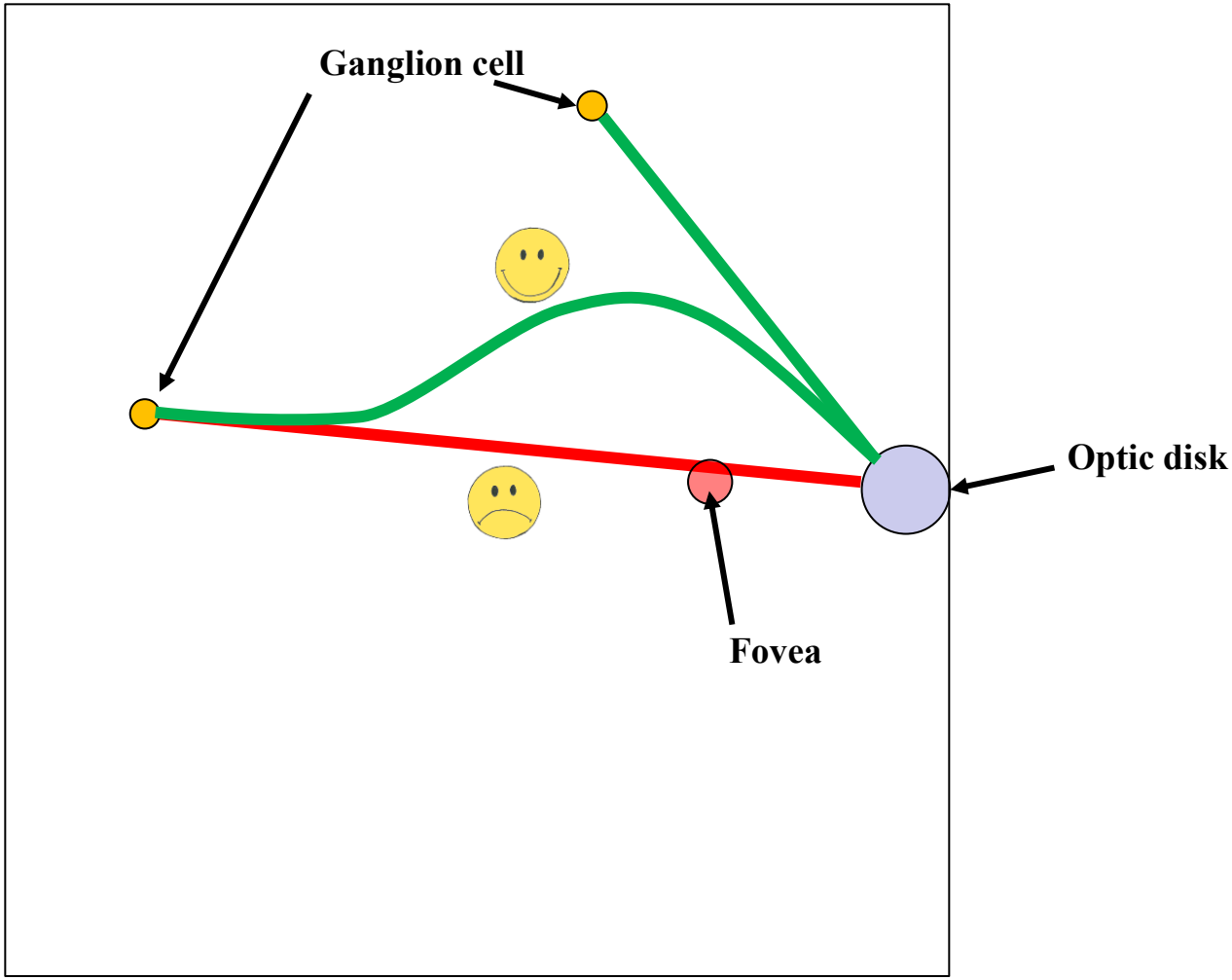


- Central Part of the macula (where most of our photopic vision occurs).
- Extremely high density of cone receptors
- Few rods. None in the foveola ($350\ \mu m \leq 1\ °$ of visual angle).
- 1.2-2 mm in diameter $\leq 2°$ of visual angle
- Displaced neural layers (hence "the pit")
- No blood vessels
- 1-1 relationship with bipolar and ganglion cells



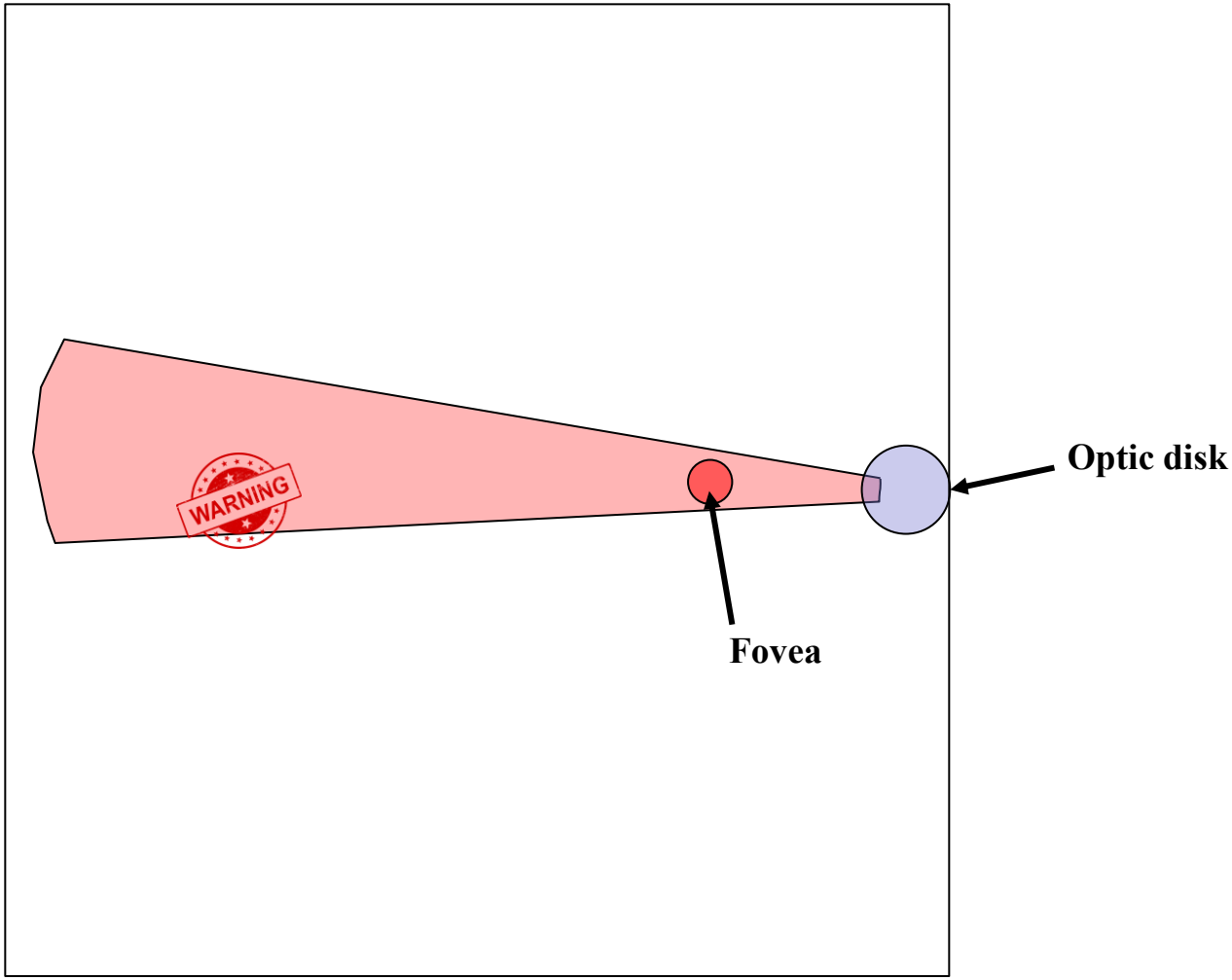
The human eye

The fovea: distribution of optic fibers



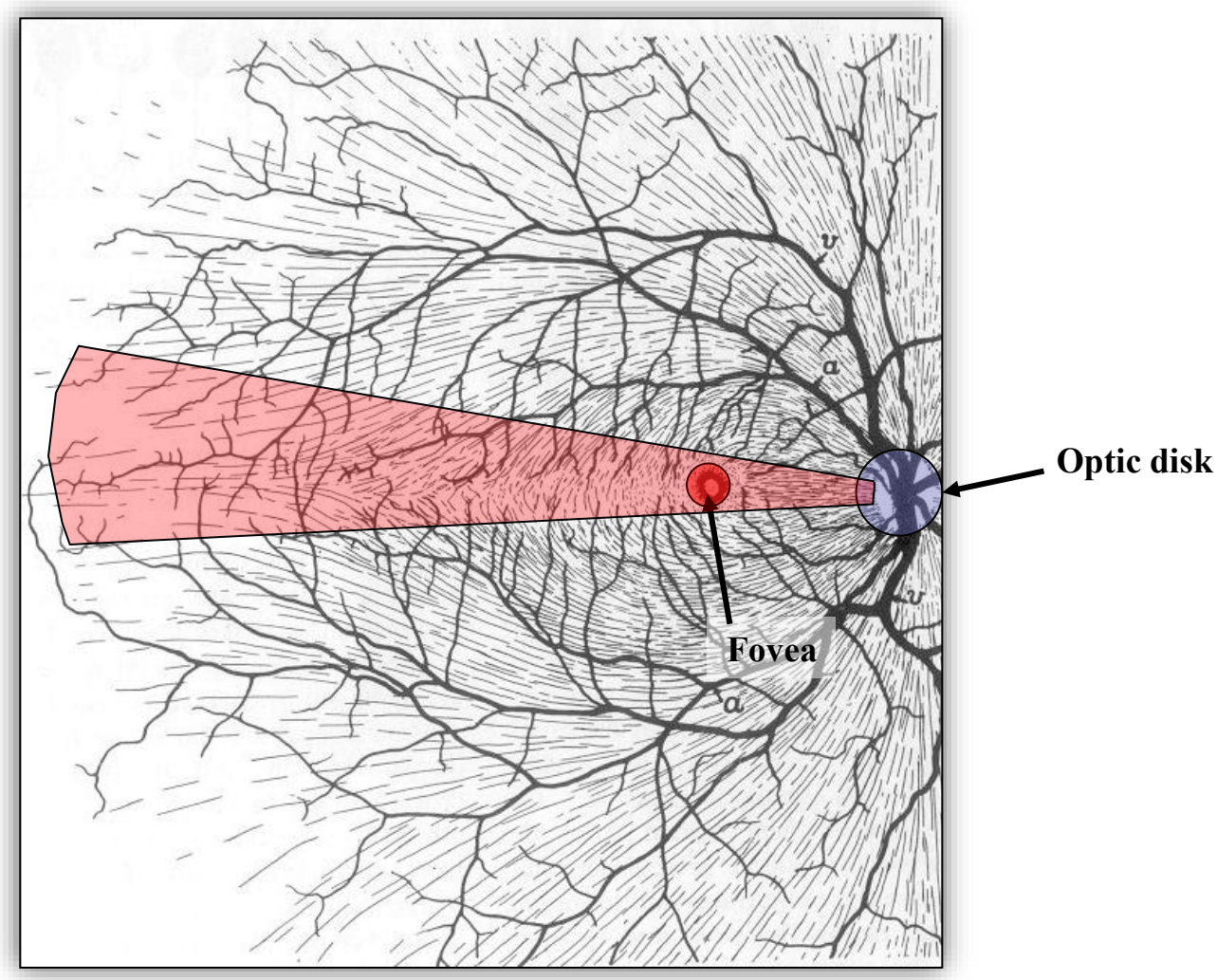
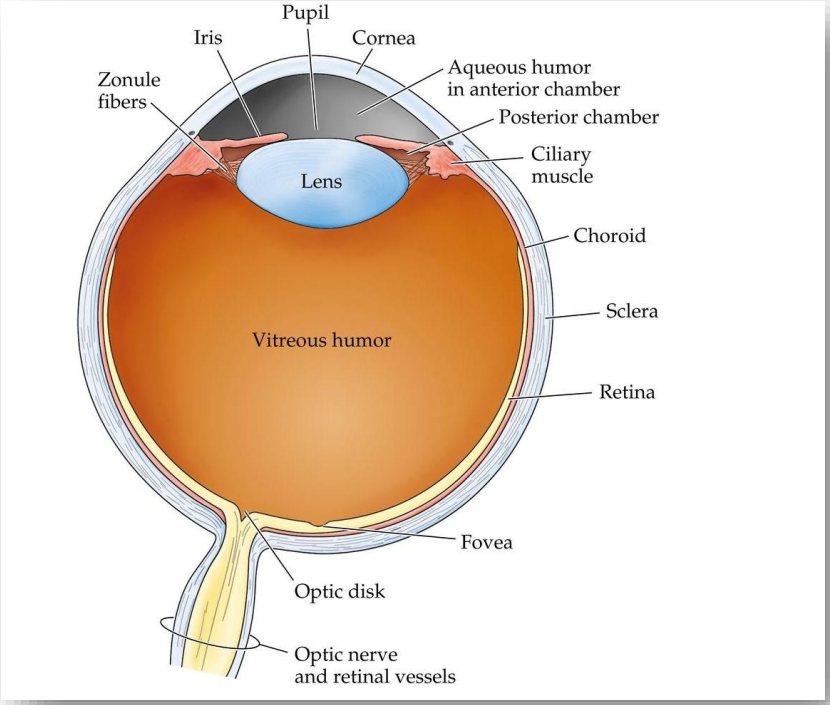
The human eye

The fovea: distribution of optic fibers



The human eye

The fovea: distribution of optic fibers



The human eye